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A comprehensive review on enhanced phase change materials (PCMs) for high-performance thermal energy storage: progress, challenges, and future perspectives

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Abstract
Latent heat thermal energy storage (LHTES) represents a promising and sustainable solution for long-term energy storage. Phase change materials (PCMs) play a crucial role in LHTES systems by effectively storing and releasing energy during phase transitions. However, their inherently low thermal conductivity (typically 0.2 to 0.7 W m^{-1} K^{-1}) restricts broader applications. This critical review explores enhancement techniques, with a particular focus on nano-enhanced PCMs (NePCMs), which incorporate nanoparticles to significantly enhance thermal conductivity (up to 300%) and improve charging/discharging rates by over 40%. The review covers the evolution of diverse methods, including the utilization of fins, geometric modifications, metal foams, and nanoparticles, to enhance heat transfer efficiency and optimize energy storage unit charging and discharging processes critical for sustainable energy solutions in thermal energy storage (TES) systems. The review consolidates experimental and numerical findings, evaluates thermophysical improvements, and outlines practical applicability. The review highlights significant advancements in thermal performance achieved through the incorporation of nanoparticles and hybrid enhancement techniques. Moreover, it underscores the urgent need for further research to address existing gaps and explore innovative designs for future thermal energy storage modules, ultimately elevating energy storage capabilities

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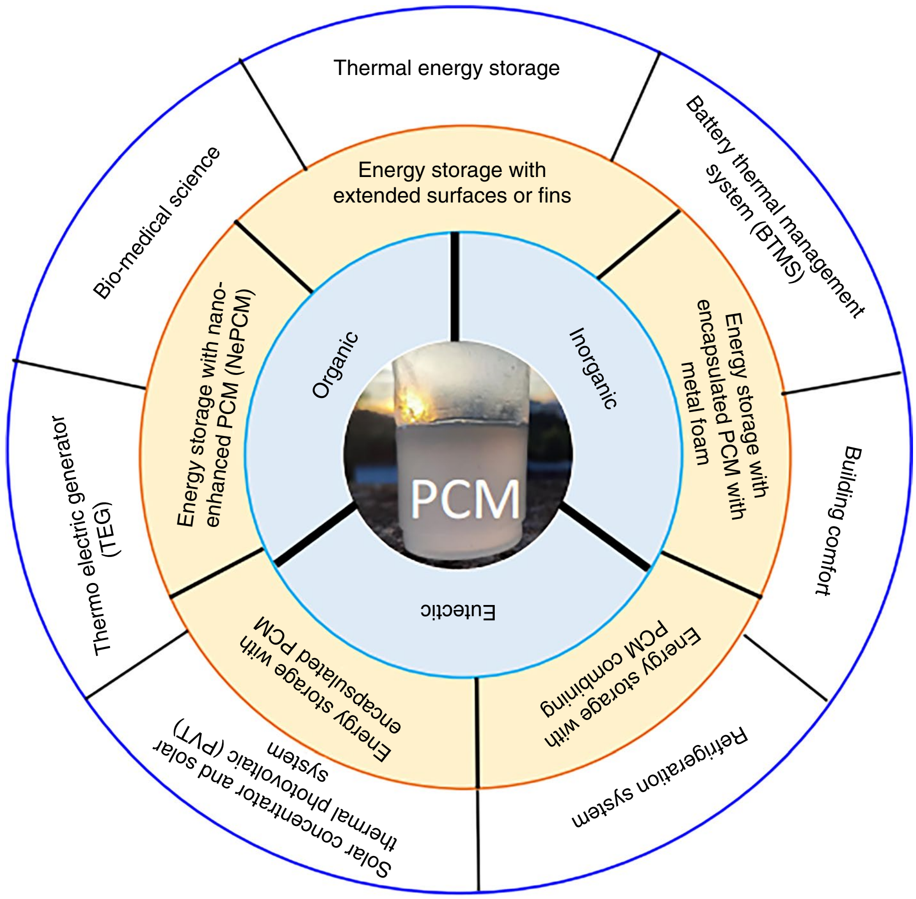
and promoting sustainability. This work offers a comprehensive framework for researchers to develop next-generation TES systems using NePCMs, targeting high energy efficiency and sustainability.

Graphical abstract



Keywords Phase change material (PCM) · Energy storage · Nanoparticles · Latent heat · Heat transfer enhancement

List of symbols

A_{mush}	Constant for mushy zone (kg m^{-3} s^{-1})
A_t	Surface area between fin and PCM (m^2)
C_p	Specific heat (J kg^{-1} K^{-1})
f	Liquid fraction
g	Gravitational acceleration (m s^{-2})
h	Sensible heat capacity (J kg^{-1} K^{-1})
k	Permeability (m^2)
K_t	Coefficient of thermal performance between fin and PCM
L	Cavity length (m)
L_H	Latent heat of melting (K J kg^{-1})
l_t	Maximum spacing between the fins (m)
L_w	Arc length (m)
Q	Energy (J)
T	Temperature (K)
t	Time (s)
u, v	x and y velocities (m s^{-1})
x, y	Cartesian coordinates (m)

Greek symbols

β	Coefficient of thermal expansion (K^{-1})
ε	Porosity
δ	Small constant number
μ	Fluid dynamic viscosity (Kg m^{-1} s^{-1})
η	Efficiency
ν	Kinematic viscosity (m^2 s^{-1})
ρ	Density (Kg m^{-3})
ψ	Dimensionless stream function
K	Thermal conductivity (W m^{-1} s^{-1})

Subscripts

m	Melting
ref	Reference
s	Solidification

Abbreviations

PCMs	Phase change materials
NePCMs	Nano-enhanced phase change materials
NPs	Nanoparticles
TES	Thermal energy storage
HP	Heat pipes
MWCNT	Multiwalled carbon nanotubes
GnP	Graphene nanoplatelets
CNF	Carbon nanofibers
SWCNT	Single-walled carbon nanotubes
CNT	Carbon nanotubes
TEG	Thermal energy generator
TMS	Thermal management system
SHS	Sensible heat storage
TCS	Thermochemical storage
HTF	Heat transfer fluid
LHTES	Latent heat thermal energy storage
M-TES	Mobilized-thermal energy storage
BTMS	Battery thermal management system
LTO	Lithium titanate
TEC	Thermoelectric cooler
TESS	Thermal energy storage systems

Introduction

With fossil fuel reserves depleting, there is an urgent need for sustainable and efficient energy storage solutions to secure long-term energy availability. In today's digital age, automation has spiked energy consumption, leading to significant energy losses (15–50%) [1]. Most lost energy is converted into waste heat, originating from sources such as air conditioning, industrial processes, batteries, engines, appliances, and exhaust. About 63% of this unused heat is below 100 °C, according to Wang et al. [2]. To address this significant energy wastage, latent heat thermal energy storage (LHTES) units [3] have emerged as a promising solution. These innovative systems employ phase change materials (PCMs) [4, 5] to capture and store the waste or unused heat as the latent heat energy. PCMs, with their varying melting points, absorb heat, transition into a liquid state, and store sensible heat energy until they are fully melted [6]. Subsequently, they release this stored heat energy when exposed to a colder environment, solidifying as heat is released through natural convection. The now-available heat can be repurposed for various applications [7, 8].

However, it is crucial to note that LHTES is just one of several thermal energy storage systems. Others include thermochemical storage (TCS) and sensible heat storage (SHS) [9]. TCS stores energy through endothermic and exothermic reactions, such as the reversible reaction of calcium hydroxide ($Ca(OH)_2$). In a chemical reaction at a working temperature of 500 °C, it exhibits an enthalpy change of 3 GJ m^{-3} [10]. The reaction is represented as $Ca(OH)_2 \leftrightarrow CaO + H_2O$. In contrast, sensible heat storage (SHS) stores energy by raising the temperature of the storage medium without a phase change and releases it as temperatures fall, typically using materials like sand, rocks, mud, water, or non-reactive metals. Water, due to its low setup costs and non-toxic properties, is a commonly employed SHS tool. However, it is important to note that while SHS is cost-effective and user-friendly, it is not as efficient as latent heat storage units [9].

PCMs are popular for energy storage due to their wide range of melting and solidification temperatures and high fusion enthalpy. They ensure nearly constant temperatures during phase transitions, unlike sensible heat storage, which undergoes temperature fluctuations. Latent heat storage maximizes energy capacity by absorbing or releasing heat energy during phase changes, without temperature variation [11]. In renewable energy, understanding energy efficiency is crucial. Thermal energy storage (TES) is key, capturing excess energy for later use. Three types of TES—sensible heat, latent heat, and thermochemical storage that offer unique benefits. Latent heat storage is particularly dense, as studies show [6]. Sensible heat and latent heat are key in thermal energy storage. Sensible heat increases the temperature without phase change, while latent heat is absorbed or released during phase changes without temperature change [6, 12, 13]. This characteristic makes them invaluable in various applications, including solar energy storage, building temperature regulation, and battery thermal management. PCMs ensure nearly constant temperatures during phase changes, offering superior energy efficiency compared to other forms of energy storage.

Despite their advantages, conventional PCMs suffer from low thermal conductivity, which limits their practical applications. To address this limitation, researchers have explored various enhancement techniques, with nano-enhanced phase change materials (NePCMs) emerging as a promising solution. By integrating nanoparticles into PCMs, NePCMs offer improved thermal properties, such as increased thermal conductivity and faster charging and discharging rates. The relationship between melting enthalpy and temperature across different PCM types has been shown in various review research paper [14]. However, there are serious lacking in the recent advancements in NePCM development, focusing on different enhancement techniques, performance metrics, and real-world applications. These gaps motivate us for planning a new review article.

This review provides a critical evaluation of advancements in NePCMs aimed at improving thermal energy storage (TES) systems. The integration of nanoparticles into PCMs is presented as a viable approach to addressing challenges like low thermal conductivity and limited energy storage capacity, which are common in traditional PCMs. The paper conducts a comparative analysis of enhancement techniques, assessing their strengths, limitations, and practical applications. Additionally, it consolidates dispersed research findings, creating a comprehensive understanding of NePCM performance while addressing critical issues such as thermal conductivity, latent heat storage, stability, and environmental impacts. By analyzing recent progress and exploring practical applications, the review identifies key trends, evaluates performance metrics, and proposes directions for future research. This work bridges theoretical advancements and real-world implementation, providing valuable insights for researchers and practitioners to develop innovative and efficient TES systems using NePCMs. Unlike other reviews, this paper specifically focuses on:

(a) Investigate the potential of LHTES units, emphasizing the role of PCMs in achieving efficient and sustainable energy storage solutions.
(b) A detailed examination of the integration of nanoparticles to improve the thermal conductivity and energy storage capacity of PCMs, highlighting cutting-edge developments in nanotechnology applications.
(c) A systematic comparison of different enhancement techniques, including the use of various additives, encapsulation methods, and composite structures, providing a clear understanding of their relative benefits and limitations.
(d) An in-depth evaluation of performance metrics such as thermal conductivity, latent heat storage capacity, thermal stability, and cyclic durability, providing a critical quantitative evaluation of recent thermal performance innovations.
(e) Insight into the practical applications of enhanced PCMs in various industries, including renewable energy systems, electronics cooling, and aerospace, with specific case studies illustrating real-world implementation and performance.
(f) Identify unresolved issues such as material compatibility, environmental sustainability, and cost-effectiveness while proposing innovative solutions.
(g) Identification of the key challenges that remain in the field, such as material compatibility, long-term stability, and cost-effectiveness, along with potential research directions and technological solutions to address these issues.

Energy storage units and their applications

A LHTES unit serves a versatile role in energy management. It functions as a heat energy storage unit by harnessing the phase transition of PCMs during the heating process [15, 16]. Beyond this primary function, it is also used as a cooling system for electronic devices [17] and lithium-ion batteries in modern electric vehicles [18], addressing issues such as car exhaust [19] and air conditioner waste heat [20]. Moreover, it plays a pivotal role in waste heat recovery systems [21, 22].

In the realm of energy efficiency, PCM-based energy storage units offer the promise of conserving energy in air-conditioning systems [23]. They capture excess heat, which can be subsequently converted into electrical energy or employed for heating purposes when the PCM undergoes solidification [24]. The integration of LHTES units into cold storage and refrigeration systems presents an opportunity to minimize power consumption and enhance comfort levels by embedding them within building walls [25]. These systems have a notable impact on elevating the efficiency of solar thermal energy storage, particularly in the context of solar photovoltaic thermal (PVT) panels, which benefit from increased cooling efficiency and electricity production. LHTES units have diverse medical applications, contributing to enhanced healthcare solutions [26, 27]. The effectiveness of PCMs extends even further when integrated with machine learning techniques, bolstering energy storage system performance. PCMs also offer a sustainable approach to address energy conservation and greenhouse gas management, making them valuable components of energy-saving strategies [28]. This multifaceted material finds application across various industries, both industrial and household areas [29–31].

Thermal energy storage (TES)

TES balances energy supply and demand for heating, cooling, and renewable energy. PCMs with high energy density are crucial. Key factors for evaluating TES include energy and power density. Balancing these ensures efficient TES performance. Researchers are exploring methods to enhance TES efficiency [32]. Microencapsulating PCM boosts thermal energy storage by improving thermal conductivity, preventing environmental interaction, and reducing leakage during melting [33]. LHTES, using PCMs for heat storage and release, effectively address energy supply–demand imbalances [34]. While renewable energy sources are essential, their intermittent nature necessitates reliable TES systems to ensure supply–demand balance [35]. Researchers have explored energy storage using organic PCMs. Rheological knowledge improves the phase change transition of PCM during melting, improving its performance [36]. Mobilized-thermal energy storage (M-TES) systems effectively absorb low-temperature industrial waste heat (IWH) using PCM [37]. For further details on TES and its effectiveness, refer to [38–40].

Battery thermal management systems (BTMS)

High-performance batteries are critical for electric vehicles (EVs), enabling extended range and rapid charging. To maintain consistent temperatures, Battery Thermal Management Systems (BTMS) are essential. PCM is a passive cooling method in cars, ensuring uniform temperatures and lower operating expenses [41–43]. PCM-based battery cooling is less complex than traditional methods like oil cooling, liquid cooling, or forced air cooling [44]. Combining PCM with biodegradable materials such as jute fiber offers an eco-friendly alternative for BTMS [45]. Innovative designs, such as active–passive BTMS with PCM and thermoelectric coolers (TECs), enhance thermal performance and temperature control. High-performance PCMs and heat pipe-based designs boost BTMS efficiency. PCM aids in the management of air-cooled batteries [46]. PCM in BTMS applications includes phase change fluid (PCF) as a heat-carrying medium and battery packs wrapped with encapsulated solid PCM [47, 48]. Novel PCMs, such as PCF and flexible phase change materials (FPCM), overcome traditional PCM constraints [49].

Building comfort and air conditioning

PCM integration in latent heat storage enhances building comfort and efficiency by stabilizing temperatures without increasing carbon emissions. By incorporating PCM into building walls, indoor thermal comfort and energy savings are optimized, ideal for energy-efficient structures [50]. Iten et al. [51] studied the PCM-based TES for the natural heating or cooling of buildings. PCM-based thermal energy storage minimizes indoor temperature variations, offering a sustainable solution for passive cooling in hot or humid areas [52]. PCM integration into building envelopes significantly reduces indoor temperatures [53]. For more details on TES and its effectiveness in ensuring building comfort, refer to [54–57]. Technological advancements have extended PCM use to space heating/cooling systems through integration with air-conditioning systems, further enhancing energy efficiency. This hybridization increases performance coefficients (~ 1.8%), achieves energy savings of approximately 5%, and reduces CO_2 emissions by nearly 30% on an annual basis [58]. PCM-integrated air-conditioning units are also applied in cold storage systems [59].

Solar concentrators and solar thermal photovoltaic (PVT) systems

Improving the efficiency of photovoltaic (PV) cells for direct electricity generation from solar energy is challenging due to the impact of rising temperatures on power generation [60]. PCM cooling is utilized to absorb excess heat in PV systems, improving both thermal and electrical stability [61].

Integrating PCMs with concentrated photovoltaic (CPV) cells enhances efficiency, lifespan, and performance by effectively cooling them, maximizing solar energy utilization [61]. Different methods like nanoparticles, heat pipes, and porous foam enhance thermal conductivity in PCMs, improving energy storage device performance [62]. Combining photovoltaic (PV) systems with thermoelectric (TE) models has shown significant effectiveness. Research reveals an average efficiency increase of 15.95% for PV cells, 17.15% for PV-PCM cells, and 17.57% for PV-PCM-TE cells annually [63].

Refrigeration systems

Thermoelectric refrigerators are cutting-edge tech that improves traditional systems, slashing energy use and cutting greenhouse gas emissions [64]. LHTES boosts refrigeration performance, using renewable energy efficiently [65].

In air-conditioning and refrigeration systems, commercially available PCMs such as RT2HC, RT5HC, OM05-P, and CrodaTherm9.5 are widely used to enhance thermal regulation and reduce energy consumption [66]. PCM shows great promise in thermal applications like ice storage, transporting temperature-sensitive materials, and preserving refrigerated trucks. Its role as a TES system makes it highly valuable [67].

Thermoelectric generators (TEGs)

In the modern world, TEGs have established themselves as clean and reliable energy sources. They offer an effective means of harnessing waste heat from various sources [19, 20]. PCMs have broad applications in various industries like solar power, automotive, aerospace, and sensors. Their integration with TEGs has notably boosted efficiency [68].

TEGs leverage daily fluctuations in ambient temperatures to generate electrical energy using PCM [69]. This sustainable energy resource has a broad range of applications, spanning from wireless sensors and solar power plants to the automotive industry and aerospace sector [68]. PCM integration improves TEG performance by regulating interfacial thermal conduction [70].

Bio-medical applications

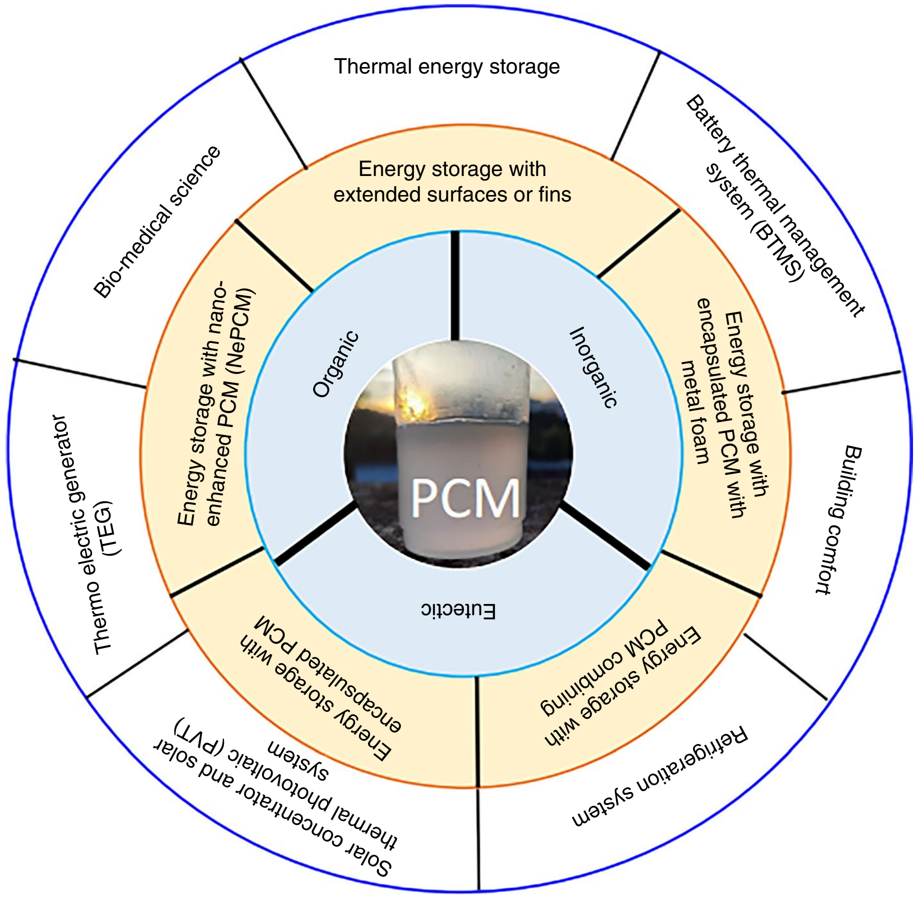
PCM also plays a very useful role in biomedical science, especially in regenerative medicine and tissue engineering. With the advancement in electronic devices, researchers such as Hirst et al. [71] have explored the use of self-assembled supra-molecular nano-structured integrated PCMs for advancing regenerative medicine. In another study, Seo et al. [72] explored metallic nano-slits, surpassing the skin-depth limit. Their research highlights the broad potential of nano-structures in advanced devices, enhancing skin-depth applications and supporting sub-nano-optics. Notably, PCM has more recently found application in tumor therapy and theranostics [73]. The reversible phase transformation of PCMs within narrow temperature ranges minimizes payload leakage and enables temperature-triggered drug release, making them ideal for precision medicine applications [74].

A summary of the diverse applications of PCMs is presented in Table 1. PCMs have been applied across a range of thermal management and energy efficiency systems. In photovoltaic (PV) cells, PCMs like PT-58 and hybrid types significantly reduce temperatures, enhancing cooling and energy output. BTMS in electric cars benefit from PCMs combined with jute or TEC, extending battery life and improving efficiency. PCMs are also used effectively in building comfort, refrigeration, air conditioning, and cold chain delivery, where they reduce indoor temperatures, improve refrigeration efficiency, and prolong cold chain thresholds. In thermoelectric generators (TEG), PCMs enhance thermal conduction, allowing power generation for internet of thinking (IoT) sensors. Solar water heaters using PCMs have shown a 10% increase in efficiency over traditional heaters, while PCM systems in air conditioning offer thermal conductivity far superior to water. PCMs contribute to efficient, cost-effective cooling and thermal management in diverse applications. A graphical presentation of diverse applications of PCMs is presented in Fig. 1.

Table 1 Summary of various PCM applications

References	Applications	PCM types	Major findings
Jamil et al. [16]	Cooling of PV cell	PT-58 nano-PCM	Nano-PCM (CB/PT-58) cools better than silica/PT-58
Jaguemont et al. [44]	BTMS	All PCM types	PCM cooling is simpler and cheaper than traditional methods
Youssef et al. [45]	BTMS in electric car	PCM combined with jute	Jute-PCM boosts thermal efficiency over standard PCM cooling
Liu et al. [46]	BTMS with PCM coupled with a TEC	PCM with a thermoelectric cooler	TEC batteries last 57.8% longer at 40 °C with a 2.5 K temperature difference
Yu et al. [47]	Electric car battery cooling	PCM with Heat pipe	PCM and heat pipes effectively control battery temperature
Qu et al. [54]	Building comfort	Paraffin wax	PCM integration decreases indoor temperature
Luo et al. [63]	PV cell	PV-PCM-TE	PV-PCM-TE system outperforms traditional PV-TE system
Bozorgi et al. [64]	PCM-thermoelectric refrigerator	RT 5HC	PCM-based TES improves food refrigeration efficiency
Longo et al. [66]	Air conditioning	Various PCMs (RT2HC, RT5HC, OM05-P, OM08, CrodaTherm9.5)	PCM thermal conductivity is 72–74% higher than water
Tuoi et al. [69]	Thermoelectric generator (TEG)	TEG with PCM	TEG powers IoT sensors and small systems
Liu et al. [70]	Enhanced TEG system performance	TEG with PCM	TEG-PCM combo boosts thermal conduction
Nabil and Mansour [75]	PV panel temperature reduction	RT28HC	Various cooling techniques boost PVT energy output
Gad et al. [76]	Thermal regulation of PV panel	Hybrid PCMs (SP31, RT35HC, RT42) with flat heat pipes	PCM integration cuts PV panel temp by 20.6 °C
Wheatley et al. [77]	Solar thermal water heater	Paraffin wax	PCM heaters are 10% more efficient than conventional ones
Burgess et al. [78]	Cold chain delivery application	PCM mixture bricks	Higher latent heat PCMs prolong the cold chain threshold

Fig. 1 Graphical presentation of diverse applications of PCMs



Types of PCMs and their characteristics

PCMs are classified based on melting points and enthalpy of fusion, and are grouped into low, medium, and high-temperature categories. Inorganic PCMs such as salts and eutectic mixtures typically exhibit high melting points (150–450 °C), while organic PCMs like paraffins and fatty acids melt between − 50 °C and 120 °C. PCMs are categorized into low (below 120 °C), medium (120–300 °C), and high (above 300 °C) melting temperatures [79, 80]. Apart from temperature-based classification, PCMs are also categorized according to their chemical composition into organic, inorganic, and eutectic materials, as shown in Fig. 2.

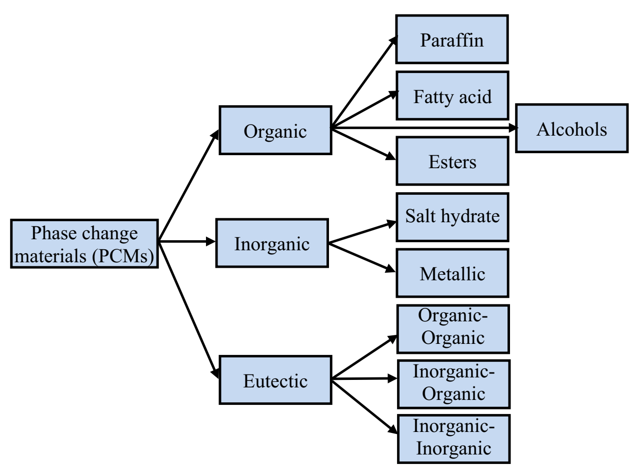


Fig. 2 Classification of various PCMs available [32]

Organic PCMs, divided into paraffin and non-paraffin types, offer versatility in thermal energy storage, as highlighted in Table 2a. Organic PCMs, such as paraffins and fatty acids, are non-toxic, non-corrosive, and exhibit stable phase change behavior with good nucleation and a broad range of melting temperatures [81]. However, they also have limitations such as flammability, incompatibility with plastics, low thermal conductivity, density issues, and instability at high temperatures due to weak covalent bonds. Although generally more expensive than salt hydrates, paraffin wax remains a popular organic PCM, alongside fatty acids, esters, alcohols, and glycols [82]. Fatty acids, although exhibiting a higher latent heat of fusion compared to paraffins, are more corrosive and expensive. Non-paraffin PCMs tend to have low flash points, limited thermal conductivity, and remain stable only at lower temperatures [83]. Inorganic PCMs, as detailed in Table 2b, include salt hydrates and metallic PCMs, which may contain salt solutions. These materials

Table 2 Different material characteristics with likely to be used as PCM

Material	Melting temperature, °C	Heat of fusion, kJ kg^{-1}	References
(a) Organic materials			
Butyl stearate	19	140	Hawes et al. [84]
Paraffin C_{16}–C_{18}	20–22	152	Zalba et al. [85]
Capric–Lauric acid	21	143	Hawes et al. [84]
Dimethyl sebacate	21	120	Feldman et al. [86]
Paraffin (RT 20)	22	172	www.rubitherm.de [87]
Polyglycol E 600	22	127.2	Dincer and Rosen [88], Lane [89]
Paraffin C_{13}–C_{24}	22–24	189	Abhat [90]
Mistiric acid (34%) + Capric acid (66%)	24	147.7	Lane [89]
Paraffin (RT 26)	25	131	www.rubitherm.de [87]
Paraffin (RT 25)	26	132	www.rubitherm.de [87]
1-Dodecanol	26	200	Hawes et al. [84]
Paraffin (RT27)	28	179	www.rubitherm.de [87]
Paraffin C_{18} (45–55%)	28	244	Abhat [90]
Paraffin (RT 30)	28	206	www.rubitherm.de [87]
Vinyl stearate	27–29	122	Feldman et al. [86]
Paraffin (RT 32)	31	130	www.rubitherm.de [87]
Capric acid	32	152.7	Dincer and Rosen [88], Lane [89]
(b) Inorganic materials			
$KF{\cdot}4H_2O$	18.5	231	Abhat [90], Naumann and Emons [91]
Salt Hydrate (Climsel C23)	23	148	www.climator.com [92]
Salt Hydrate (Climsel C24)	24	216	www.climator.com [92]
$Mn(NO_3)_2{\cdot}6H_2O$	25.8	125.9	Wada et al. [93]
Salt Hydrate (S27)	27	207	www.cristopia.com [94]
$CaCl_2{\cdot}6H_2O$	29	190.8	Dincer and Rosen [88], Lane [89]
Salt Hydrate (TH 29)	29	188	www.teappcm.com [95]
$Na_2SO_4{\cdot}10H_2O$	32	251	Hawes et al. [84], Abhat [90], Naumann and Emons [91]
Salt Hydrate (Climsel C32)	32	212	www.climator.com [92]
(c) Eutectic PCMs			
0.4% Cl + 4.3%NaCl + 48%$CaCl_2$ + 47.3% H_2O	26.8	188	Abhat [90]
53% $Mg(NO_3)_2{\cdot}6H_2O$ + 47%$Ca(NO_3)_2{\cdot}4H_2O$	30	136	Abhat [90]
40% $CO(NH_2)_2$ + 60%$Na(CH_3COO){\cdot}3H_2O$	30	200.5	Li et al. [96]

offer broader melting temperatures, higher heat of fusion, and better thermal conductivity than organic PCMs. However, they are prone to thermal instability, corrosion, and issues such as supercooling and phase separation, limiting their suitability for LHTES applications. Salt hydrates consist of solid crystals formed from inorganic salts and water, while metallic PCMs boast high thermal conductivity and specific heat, though they are more expensive than organic alternatives [32, 83]. Eutectic PCMs, discussed in Table 2c, represent a newer class of PCMs known for their adjustable melting temperatures. Comprising combinations of two or more components, eutectic PCMs offer tunable melting temperatures based on compositional ratios, making them ideal for tailored thermal storage applications. These materials offer high latent heat of fusion and good thermal conductivity but are costly, despite their promising potential [9].

Various PCMs are available in the market, offering a range of prices and properties, each with unique thermal and chemical characteristics, making the selection process crucial for specific applications. Choosing the right PCM depends on thermophysical, chemical, and economic considerations for the intended use. Figure 3 presents a flowchart

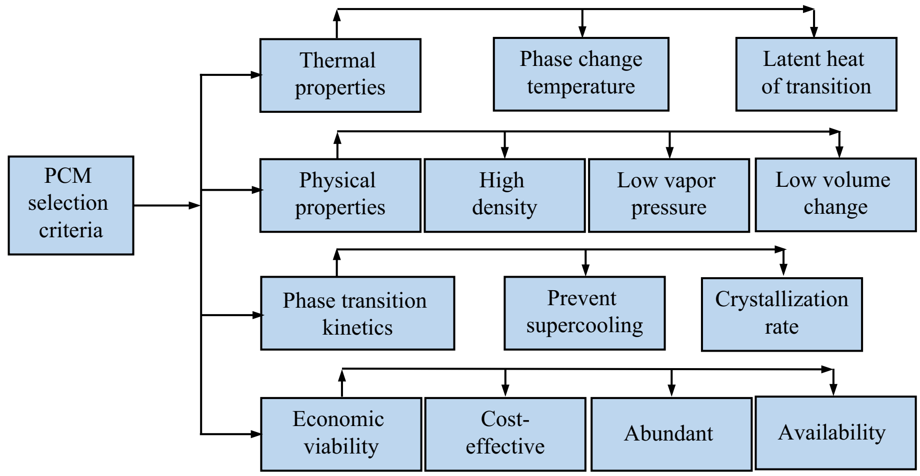


Fig. 3 Flow chart of desirable properties of latent heat energy storage material

that outlines the process for selecting the best PCM based on specific application requirements [7]. An ideal PCM should exhibit high specific heat, significant latent heat of fusion, a narrow melting/freezing temperature range, and excellent thermal conductivity in both its liquid and solid phases. Additionally, during phase transitions, it should have minimal volume change and low vapor pressure, as highlighted by key properties such as high latent heat of fusion, a narrow melting/freezing range, and good thermal conductivity [97]. Mechanical integrity is essential; selected PCMs must be non-toxic, non-flammable, and environmentally benign, ensuring long-term safety in thermal applications. The kinetics of phase transitions are essential; fast transitions enable efficient heat transfer, and ideal PCMs have low hysteresis with minimal supercooling or superheating, melting or solidifying precisely at the desired temperature without needing excess energy. Chemically, PCMs must be non-toxic, non-corrosive, non-combustible, and non-explosive, with chemical stability and the ability to reversibly melt and solidify without degradation over numerous cycles. They should also minimize sub-cooling during solidification and be compatible with the materials they contact [32]. Finally, the economic viability of PCMs is crucial, taking into account factors such as cost, availability, ease of manufacturing, and recyclability. Economically, PCMs should be cost-effective, readily available, and recyclable—ensuring that their lifecycle benefits justify initial deployment costs.

Enhancing heat transfer in PCM-based systems

Due to their inherently low thermal conductivity, conventional PCMs exhibit limited heat transfer performance. To overcome this, several passive enhancement techniques—such as the incorporation of extended surfaces (fins), metal foams, high-conductivity nanoparticles, and porous media—have been developed to accelerate the charging/discharging process and improve energy efficiency in LHTES systems [98–102]. Numerous research studies are published annually, focusing on improving the thermal conductivity and energy performance of PCMs through both experimental and numerical approaches. Researchers focus on various PCMs and methods to improve heat transfer. PCMs are crucial for energy-efficient devices, storing and releasing significant heat during phase transitions. There is a continuously growing interest in enhancing heat transfer in thermal energy storage units. PCMs struggle with slow melting and freezing due to poor thermal conductivity. Researchers are addressing this by developing passive methods to boost thermal conductivity, speed up phase changes, and improve the efficiency of latent heat storage systems without extra energy input.

During the melting and solidification process, heat transfer within PCMs primarily occurs through conduction. However, convection currents arise in the liquid phase, and the addition of thermally conductive structures or materials facilitates faster heat propagation. Examples of passive enhancement techniques include using fins, heat pipes, metallic foams, and dispersing conductive nanoparticles in PCMs [103]. Optimizing fin size in a triplex tube-type heat exchanger with pure PCM [104] enhances heat transfer. Carbon and metal-based PCMs offer superior thermal properties, promising better performance [97, 105]. System orientation, temperature, and fluid flow rate have also been extensively studied in experiments. In general, heat energy storage and subsequent release follow a typical heating and cooling cycle of PCM. During the day, solid PCM absorbs solar energy, melts, and stores thermal energy. At night, it releases the stored energy as it solidifies. However, PCMs melt and freeze slowly due to their low thermal conductivity.

Passive heat transfer enhancement methods improve PCM-based TES systems by using techniques like enhanced conduction materials and optimized heat exchanger designs, aiming for more efficient energy storage and utilization. Research in this field is ongoing to maximize these benefits.

Energy storage with extended surfaces or fins

Improving LHTES systems can be efficiently achieved by increasing the surface contact area between the PCM and the heated surface. This method is essential for developing cost-effective LHTES systems that ensure adequate energy density and fast thermal response times [106]. Kamkari et al. [107] found that using aluminum fins with graphite is effective for high-temperature systems. They studied how attaching fins to a rectangular container at different angles affects the charging process of lauric acid PCM. The study looked at container orientations: vertical (90°), 45°, and horizontal (0°), with and without fins heated from the side. The study found that reducing the enclosure inclination angle (Θ) from vertical to horizontal significantly increased the PCM charging rate for both finned and non-finned containers (Fig. 4a). The flat enclosure without fins improved heat transfer by 56%, while the enclosure with 3 perpendicular fins showed a 115% enhancement compared to the vertical enclosure with fins. This highlights how the orientation of the enclosure significantly impacts the charging process. Among them, the horizontal container with 3 fins performed the best, leading to the shortest charging time.

Tiari et al. [108] analyzed LHTES systems with finned heat pipes. They found that more heat pipes increased the melting rate but lowered the heated wall temperature. Longer fins reduced temperature differences within the PCM-filled chamber. Hosseinizadeh et al. [109] studied a PCM-based

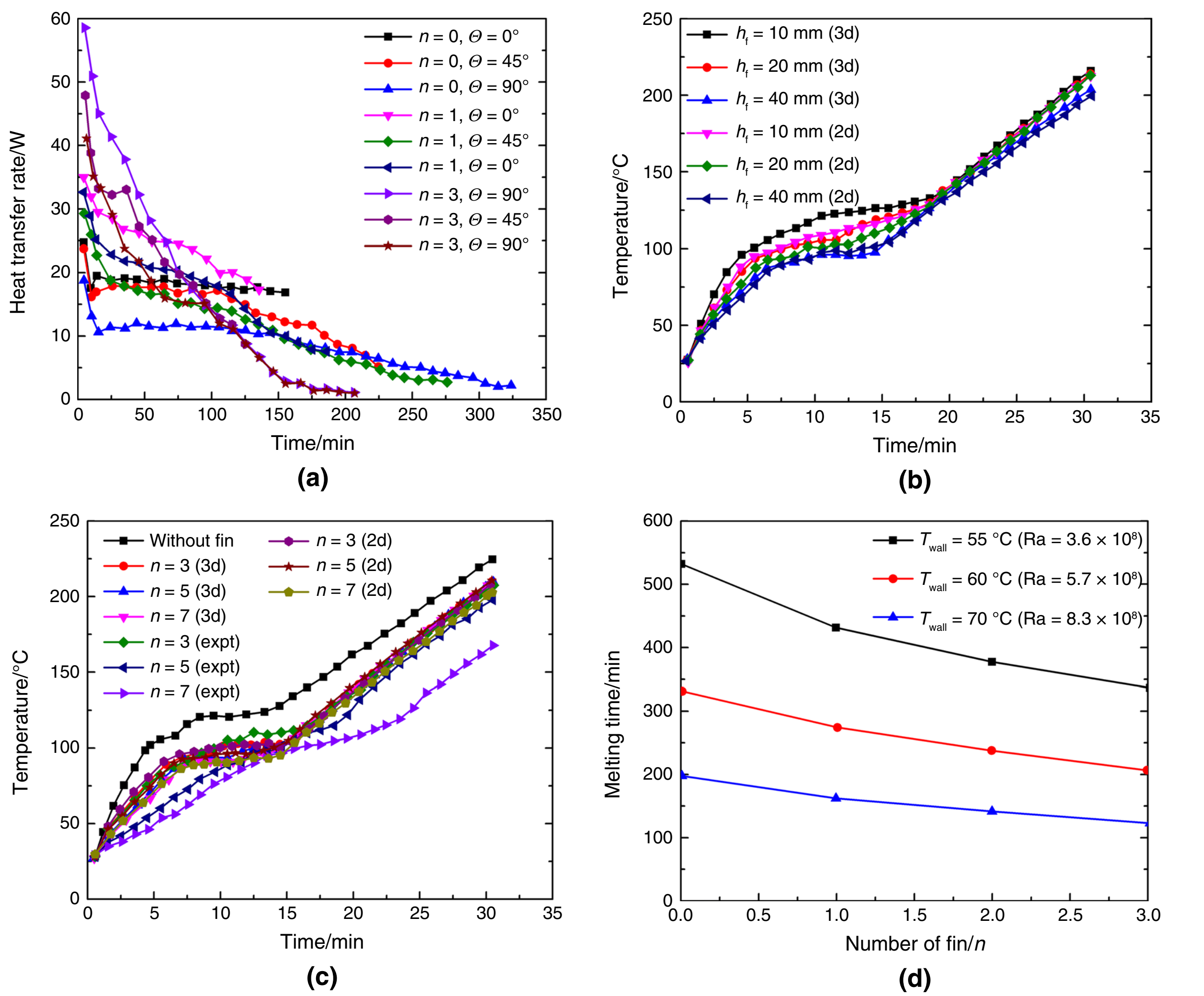


Fig. 4 Variation of **a** heat transfer with time for the different combinations of container inclinations (Θ) and number of fins (n) [107], **b** temperature with time steps for different heights of fin (h_f), and **c** temperature with time steps for different numbers of fin (n) [109], and **d** total melting time with respect to number of fins (n) at different wall temperatures [116]

heat sink and the effect on thermal system regulation. They experimented with power levels, fin numbers, height, and width, finding improved thermal performance with increased fin height (Fig. 4b) and numbers (Fig. 4c). As melting started, convection dominated heat transport, transitioning to free convection, speeding up PCM melting.

Fekadu et al. [110] studied the melting rates of PCM in a rectangular container with three adiabatic sides and one isothermal side, using an angled fin. They concluded that adjusting the number and orientation of fins can enhance the effectiveness of the system and reduce PCM melting time. Tiari et al. [111] conducted a numerical analysis on the effect of annular fins on LHTES efficiency during melting and freezing. They found that uniform fin arrangement reduced charging and discharging time by 76.3%. Wu et al. [112] designed an LHTES module with spider web-shaped fins, optimizing the solidification rate using an 8-branch design. This module significantly reduced solidification time and achieved a 1.44 times higher time-averaged heat-release rate compared to plate fins. Wu et al. [113] conducted simulations on LHTES units, finding that unequal-length compensating fins reduced charging and discharging times by 64.2% and 82.1%, respectively. These fins also decreased the heat storage-release duration by 73.3% and increased the heat transfer rate by 155%, significantly enhancing the thermal effectiveness and flexibility of the unit. Luo et al. [114] found that a combined fractal fin heat exchanger reduced PCM melting time by 68% compared to a traditional fractal fin heat exchanger. Abidi et al. [115] numerically investigated the melting and solidification of PCM in a triplex tube heat exchanger with internal and external fins. They found that an 8-cell PCM unit solidified 35% faster than a unit with 1 mm thick fins. Kamkari et al. [116] experimented with PCM-filled horizontal containers, some with fins, in thermally insulated enclosures. One side was heated isothermally at 55, 60, and 70 °C. They found that more fins reduced melting time, as shown in Fig. 4d.

Increasing the number of fins reduces the melting time and increases average speed of heat transfer rate. More fins lead to higher melting time enhancement and total fin efficacy, particularly beneficial in initial melting stages. Partial fin arrangements are advantageous, especially with higher wall temperatures. Liu et al. [117] discovered that adding triangular fins in a longitudinal arrangement to the tube and shell-type LHTES greatly enhanced freezing performance. Arıcı et al. [118] found that adding a fin to a square container with PCM reduced charging time by 13–68% depending on fin placement and 27–52% based on fin length in their numerical study.

Groulx et al. [119] numerically studied PCM charging in a container with fins and adjustable inclination. They found that wider fins connecting the front and back sides improved the thermal control of the PV-PCM module. Sciacovelli et al. [120] found that tree-shaped fins boosted tube-and-shell LHTES efficiency by 24% numerically. The best design varied based on the working duration of LHTES unit. Zhao et al. [121] conducted a numerical analysis of PCM within a rectangular container heated isothermally. They identified fin spacing, thickness, and heat transmission ratio as key factors affecting optimal fin length, particularly when natural convection was minimized. Rahimi et al. [122] experimentally tested a heat exchange unit with aluminum fins and copper tubes. Figure 5a features a unit with a transparent plexiglass box, enabling the visualization of PCM distribution among fins and tubes. They observed that using fins raised the average fin temperature consistently across different flow regimes (Fig. 5b). Changing the fin pitch didn't affect performance much. Also, increasing the input temperature from 50 °C to 60 °C decreased melting time significantly compared to an increase from 60 °C to 70 °C.

Joneidi et al. [124] conducted an experiment with a cube where all sides are insulated. Heat flux warms the bottom plate. Plate fins improve performance. The Plexiglas container, 20 mm thick, is 100 mm wide, 50 mm high, and 30 mm deep inside. A low-conductive translucent film acts as an insulator and allows for photography. Figure 6a shows the experimental setup with a thin Mica layer (0.2 mm thick) between the heater and copper heating plate. The setup includes a data logger, computer, and 36 K-type thermocouples for temperature measurement. Among these, 32 thermocouples measure PCM temperature, 3 measure heating plate temperature, and 1 measure ambient temperature. Fins are attached to vertical rows of thermocouples, with Paraffin RT35 from Rubitherm as the PCM and copper fins.

The effects of fin numbers (Fig. 6b), height (Fig. 6c), and thickness (Fig. 6d) can be categorized into three groups. Research findings indicate that as these parameters increase, the time for phase change from solid to liquid decreases. This is due to enhanced surface area, accelerating heat transfer according to Fourier law, thereby speeding up the melting process. Additionally, integrating fins into the enclosure reduces PCM mass, further expediting melting.

The PCM was melted and then poured into the enclosure, where it subsequently solidified upon contact with the sidewalls, trapping air pockets. Research shows more fins improve temperature uniformity, aiding phase change. The number of fins greatly affects heat-sink base thermal management, especially at lower temperatures. Nonlinear regression analysis establishes relationships between PCM melt percentage and base temperature. Jmal et al. [125] numerically studied paraffin C18 solidification in a heat exchanger with coaxial tubes, fins, and air passages. They found that reducing PCM liquid gaps affects thermoconvective flow growth more than enhancing solidification, suggesting fins development isn't crucial for improving the process. Ji et al. [126] numerically studied PCM melting in a TES

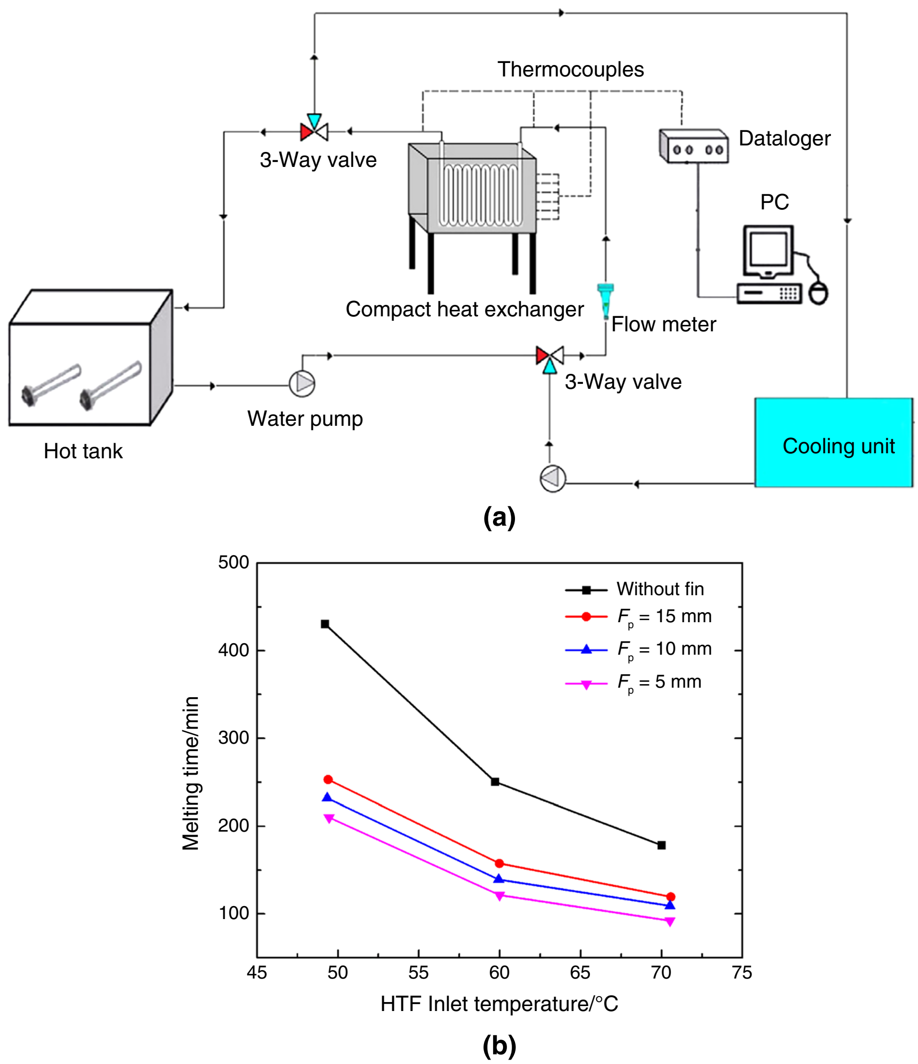


Fig. 5 (a) Experimental setup of tube and fin type heat exchanger comprising PCM [122], (b) variation of melting time at different inlet temperatures of HTF [123], where the symbol 'F_p' corresponds to the fin pitch

with a rectangular container. They found that fins arranged downward at 15° accelerate melting. Other researchers have explored using fins to improve LHTES, summarized in Table 3, which includes thermodynamic properties of PCM and fin, along with study summaries.

Several studies have explored the use of PCMs with extended surfaces or fins to enhance thermal performance. For high-temperature applications, Steinmann et al. [106] found that graphite-coated aluminum fins in $NaNO_3$ systems improved heat transfer. Kamkari et al. [107] demonstrated that changing the orientation of lauric acid without fins could improve melting rates compared to vertical fin arrangements. Tiari et al. [108] showed that fin design in KNO_3 systems had a significant impact on thermal performance, while Hosseinizadeh et al. [109] reported that the efficiency of paraffin wax (RT-80) systems improved with an increased number, height, and thickness of aluminum fins. Fekadu et al. [110] and Wu et al. [112] found that lauric acid PCMs with spider-web and fractal fins reduced solidification and melting times significantly, and Liu et al. [117] showed that triangular fins were more effective than rectangular fins for Li_2CO_3-K_2CO_3-Na_2CO_3. Other studies, such as those by Zhao et al. [121] and Rahimi et al. [122], emphasized the importance of fin spacing, length, and flow rate in optimizing PCM performance. Overall, the use of fins in PCM systems, particularly in configurations like rectangular, annular, and bifurcating designs, demonstrated improvements in thermal storage and heat transfer across different PCM types, with material choices like aluminum and copper playing a crucial role in performance enhancement. The high thermal conductivity of metal foams allows rapid heat conduction pathways within the PCM matrix, reducing thermal gradients and promoting uniform melting. Additionally, pore architecture influences convective enhancement and latent heat recovery.

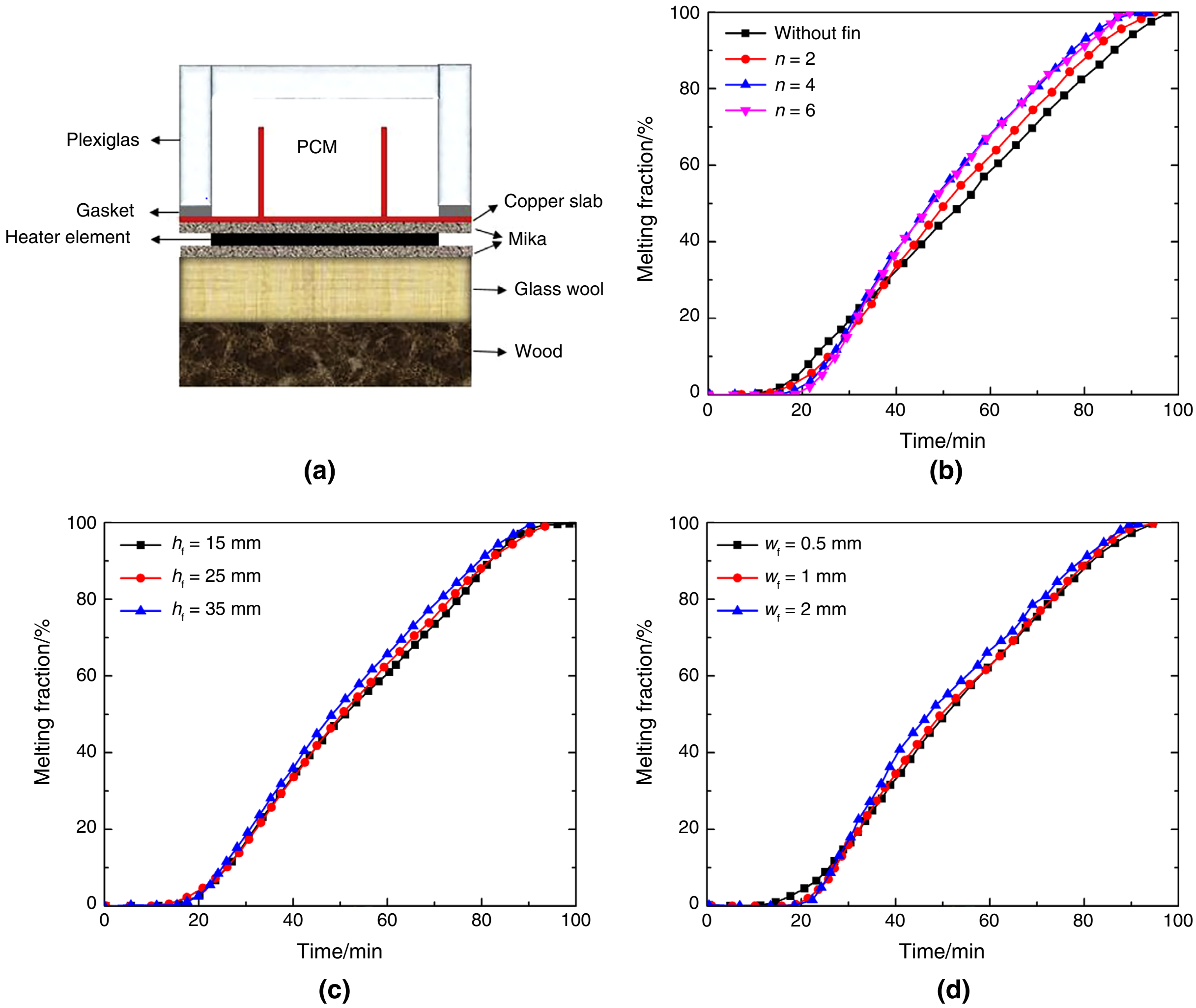


Fig. 6 **a** Experimental setup for melting phenomena of PCM-filled horizontal enclosure fitted with fins, **b** impact of increasing the fin numbers (n), **c** impact of increasing the fin height (h_f), and **d** impact of increasing the fin thickness (w_f) [124]

Energy storage with encapsulated PCM with metal foam

Extensive research has focused on enhancing thermal energy storage using PCM encapsulation, both with and without metal foam. Parida et al. [127] numerically analyzed PCM melting in metal foam composites and observed that increased porosity and larger pore diameters enhance thermal convection and reduce overall melting time. Deng et al. [128] experimentally found that using porous metal foam with a smaller fractal diameter in a PCM-porous metal foam system enhances the melting process and increases heat transfer rates. In a numerical study, Dinesh et al. [129] found that inserting metal foam with smaller pores into a PCM container consistently enhanced the energy storage during the melting process. Baruah et al. [130] studied a hybrid system with metal foam and PCM, finding that metal foam reduced PCM melting times. Smaller capsules, lower porosity, and thicker shells further decreased melting periods. Jiao et al. [131] numerically studied a PCM/metal foam hybrid energy storage system and recommended a revised mixture theory framework.

Esapour et al. [132] used computational methods to study phase change in a container with metallic porous foam. They found that adding metallic foam and more inner tubes greatly accelerated melting and solidification rates. Combining PCM with metallic foam led to faster phase change compared to pure PCM. Introducing metal foam with porosities of 0.9 and 0.7 reduced melting time by 14% and 55%, respectively. Huang et al. [133] studied how different incline angles affect the melting of paraffin in a PCM container with copper foam. They discovered that lower pore density and higher Rayleigh numbers in the metal foam speed up the melting process. Sardari et al. [134] studied radiator-based compact latent heat storage (CLHS) for home heating. They discovered that higher metal foam permeability decreased melting

Table 3 A summary of studies that used PCM with extended surface or fin

References	Used PCM	Property of PCM							Type of fins	Fin material	Case study	Important findings
		ρ /kg m^{-3}	C_p kJ kg^{-1} K^{-1}	μ /Pa s	k W mK^{-1}	L_H/kJ kg^{-1}	β /K	T_m /°C				
Steinmann et al. [106]	$NaNO_3$	1900	–	–	–	175	–	306	Circular	Aluminum and Graphite	Numerical	Applications requiring high temperatures might make use of graphite-coated Aluminum fins
Kamkari et al. [107]	Lauric acid	940	2.18	6.298×10^{-3}	0.16	187.21	–	48.2	Rectangular	–	Experimental and Numerical	Melting rate accelerates more effectively by lowering geometry orientation without fins compared to a vertical arrangement with fins
Tiari et al. [108]	KNO_3	2109	0.953	2.59×10^{-3}	0.5	95	2×10^{-4}	335	Rectangular	Nickel	Numerical	Fin design significantly affects overall thermal performance, while fin numbers have minimal influence on PCM
Hosseinizadeh et al. [109]	Paraffin wax (RT-80)	920	2.40	8.648×10^{-3}	0.2	175	–	81	Rectangular	Aluminum	Experimental and Numerical	Thermal behavior of finned heat sink improves with an increase in the number, height, and thickness of fins
Fekadu et al. [110]	Lauric acid	912.5	2.285	6.113×10^{-3}	0.15	187.21	5×10^{-4}	316.65–321.35	Horizontal rectangular	Aluminum	Numerical	Melting time increases with increased fin number and inclination, dropping rapidly at 45 degrees
Tiari et al. [111]	Rubitherm RT55	880/770	2.00	30×10^{-3}	0.2	172 ± 7.5%	1.10×10^{-4}	324–330	Annular fins (linear, tapering)	Copper	Numerical	Uniform fin arrangement reduces overall charging and discharging time by 76.3% compared to varying lengths

Table 3 (continued)

References	Used PCM	Property of PCM							Type of fins	Fin material	Case study	Important findings
		ρ /kg m^{-3}	C_p kJ kg^{-1} K^{-1}	μ /Pa s	k W mK^{-1}	L_H/kJ kg^{-1}	β /K	T_m /°C				
Wu et al. [112]	Lauric acid	862	2.30	–	0.147	178	6.15×10^{-4}	318	Spider web and plate	Aluminum 6061	Numerical	Spider-web fins reduce solidification time by 47.9% compared to plate fins
Wu et al. [113]	Lauric acid	885	2.18/2.39	–	0.16/0.14	187.21	9.25×10^{-4}	316.5/321.2	Compensating rectangular	Aluminum	Numerical	Fins reduce charging and discharging rates by 64.2% and 82.1%, respectively, while enhancing heat transfer by up to 155%
Luo et al. [114]	Lauric acid	1000	2.15/2.203	–	.15/0.22	178	–	315/317	Fractal with different bifurcation	Aluminum	Numerical	Coupled fractal fins reduce melting time by 68% compared to conventional fractal fins
Al-Abidi et al. [115]	RT82 (Rubitherm GmbH-Germany)	950/770	2.00	34.99×10^{-3}	0.2	176,000	10×10^{-4}	350.15–358.15	Longitudinal external and internal fins	Copper	Numerical	Fin length impacts thermal behavior more than fin thickness
Kamkari et al. [116]	Lauric acid	940/885	2.180/2.39	6.3×10^{-3}	0.16/0.14	187.21	–	316.5–321.5	Horizontal partial	Aluminum	Experimental	Melting period shortens with more fins, enhancing overall heat transfer
Liu et al. [117]	Li_2CO_3-K_2CO_3-Na_2CO_3	2310	1.540/1.64	–	1.69/1.60	–	–	668	Traditional rectangular and longitudinal triangular	Aluminum	Numerical	Triangular fins solidify PCM 38.30% faster than rectangular fins
Arıcı et al. [118]	Paraffin wax	750	2.89	0.001 exp (−4.25 + 1790 T)	0.21/0.12	173.4	–	319.15/321.15	Internal	–	Numerical	Comparable thermal performance to long fins achieved using nanoparticles and shorter fins
Groulx et al. [119]	RT25	785/749	1.80/2.40	1.798×10^{-3}	0.19/0.18	232	10×10^{-4}	299.75	Inclined	Aluminum	Numerical	Wider fin leads to effective management of PV-PCM module

Table 3 (continued)

References	Used PCM	Property of PCM							Type of fins	Fin material	Case study	Important findings
		ρ /kg m^{-3}	C_p kJ kg^{-1} K^{-1}	μ /Pa s	k W mK^{-1}	L_H/kJ kg^{-1}	β /K	T_m /°C				
Sciacovelli et al. [120]	Paraffin wax	812.5	2.30/1.80	–	0.2	230	–	328–333	Y- shape single bifurcation and double bifurcation	–	Numerical	Short operation periods favor large-angled Y-shaped fins
Zhao et al. [121]	lauric acid	940/885	2.18/2.39	5.9×10^{-3}	0.16/0.14	187,210	8×10^{-4}	316.65–321.35	Vertical straight	Aluminum and Stainless-Steel	Numerical	Ideal fin length is influenced by fin spacing, thickness, and heat conductivity ratio when natural convection is minimal
Rahimi et al. [122]	RT35	860/770	2.00	–	0.2	170	6×10^{-4}	302/308	–	Aluminum	Experimental	Increased flow rate reduces solidification times more significantly than melting times by fin
Bouzennada et al. [123]	Rubitherm GmbH; RT27	760/870	2.40/1.80	3.42×10^{-3}	0.15/0.24	179	5×10^{-4}	302.15	Mid-separating with varying inclination angles	Aluminum	Numerical	Fins reduce melting time by 1.28% to 20.52%, enhancing thermal energy storage rate by 14.8% to 36.9%
Joneidi et al. [124]	Paraffin RT35	860/770	2.00	23×10^{-3}	0.2	170	–	302/308	Plate	Copper	Experimental	PCM melting time shortened by increasing the number or length of fins
Jmal et al. [125]	Paraffin C18	814/774	2.15/2.18	1.2×10^{-3}	0.15/0.148	244	8.5×10^{-4}	301	Horizontal	Aluminum	Numerical	Limited PCM liquid gaps limit the impact of growing fins on improving the solidification process
Ji et al. [126]	RT42	880/760	2.00	23.5×10^{-3}	0.2	165	1×10^{-4}	315	Inclined (0,15,30,– 15,– 30)	–	Numerical	Fin with a 15° slope slightly outperforms traditional rectangular fin with reduced heat flux input

times by 95%, favoring systems with greater porosity for stable temperatures. Iasiello et al. [135] explored PCM-filled aluminum box housings ($50 \times 50 \times 50$ mm) cooled by a Peltier module connected to a liquid cooling loop. They used a 0.96 to 1% filling ratio of liquid paraffin wax in aluminum foam. Using an IR camera and MATLAB, they tracked the position of melting front over time-based on temperature data, noting that lower porosity foams resulted in quicker melting times. Mahdi et al. [136] analyzed melting and freezing in metal foam with PCM, finding that higher porosity led to faster PCM melting and lower temperatures with increased airflow.

PCM integrated with metal foam exhibit significant improvements in thermal conductivity and energy storage efficiency due to the high conductivity of the metal matrix. From the studies summarized in Table 4, it is evident that aluminum is the most commonly used metal foam due to its relatively high thermal conductivity and lightweight nature. For instance, Parida et al. [127] and Baruah et al. [130] reported that the porosity of foam significantly affects the melting process, with lower porosity leading to faster melting but less latent energy storage capacity. Conversely, higher porosity foams, while slower to melt, can hold more latent energy, which is beneficial for applications requiring prolonged heat storage. Additionally, studies such as Deng et al. [128] highlighted that the geometry of metal foam and pore size impact not just the melting time but also the system's temperature sensitivity, suggesting that customizing foam structures can optimize thermal performance for specific applications. Although copper and nickel foams have been explored, as seen in the work by Jiao et al. [131], aluminum foams remain more popular due to their cost-effectiveness and manageable thermal properties, indicating a trade-off between performance and economic feasibility.

Energy storage with PCM combining heat pipes

Heat pipes act as efficient thermal conductors due to phase-change-based internal fluid cycling, enabling rapid heat redistribution. When embedded into PCM systems, they significantly accelerate both charging and discharging phases. Research has explored using heat pipes to improve the effectiveness of LHTES. Sharifi et al. [137] found that increasing heat pipe diameter and condenser length enhances melting rates in a vertical cylinder container filled with PCM, through numerical examination of isothermal heating strategy. Sharif et al. [138] used a heat pipe-metal foil setup to study PCM phase change. They found a 300% faster process with heat pipes and metal foil, reducing thermal resistance. Xiaohong et al. [139] numerically studied high-temperature PCM for energy storage with a heat pipe receiver, finding significant temperature variations in the PCM due to void cavities, which could impact the longevity of the receiver.

Shabgard et al. [140] found that integrating heat pipes into a high-temperature LHTES module enhanced its thermal efficiency for solar thermal electricity generation. Peng et al. [141] created a numerical model for a cylindrical lithium-ion battery with PCM and a heat pipe hybrid cooling system, revealing uneven PCM liquid distribution during discharge. Tiari et al. [142] studied PCM-based TES with heat pipes, finding varied charging times influenced by flow rate and HTF temperature. Pawar et al. [143] showed that their heat pipe-based TES system, combined with PCM, achieved a maximum temperature difference of 30 °C during nighttime peak hours. Chen et al. [144] simulated a battery thermal management system with PCM and heat pipes, discovering that lower thermal conductivity in the heat pipes led to improved PCM melting. Behi et al. [145] achieved 86.7% cooling capacity in electronic cooling appliances using a PCM-assisted heat pipe module. Behi et al. [146] studied electric vehicle thermal management with PCM, heat pipes, and free convection. They found heat pipe-assisted PCM modules significantly decreased cell temperature compared to natural convection.

PCM with heat pipes show considerable potential for enhancing thermal management by facilitating rapid heat distribution and reducing temperature gradients. Table 5 indicates that integrating heat pipes with PCMs leads to a marked increase in melting rates and overall system performance. Sharif et al. [137] observed that using sodium nitrate in conjunction with stainless steel heat pipes significantly increases melting rates, especially with variations in heat pipe diameter and condenser length. Furthermore, Sharif et al. [138] demonstrated that n-octadecane-based PCM systems equipped with heat pipes can achieve melting rates approximately nine times higher than those of bare copper rods, highlighting the efficiency of heat pipe integration. Meanwhile, studies such as Behi et al. [145] emphasize that PCM-assisted heat pipes can cover up to 86.7% of the cooling load in power systems, showing their effectiveness in applications requiring rapid heat dissipation. However, the effectiveness of these systems can vary depending on factors such as PCM thickness distribution, heat pipe material, and the operating conditions, as noted by Chen et al. [144]. This variation suggests that while heat pipe integration offers enhanced thermal performance, optimizing the design parameters is crucial for maximizing their benefits across different use cases.

Energy storage with encapsulated PCM

Encapsulated PCM finds applications in various fields for thermal energy storage and release. Tittelein et al. [147]

Table 4 A summary of studies of PCM with metal foam

References	Used PCM	Property of PCM							Property of metal foam/ Heat pipe/ Nano-particle				Case study	Important findings
		ρ /kg m^{-3}	C_p /kJ kg^{-1} K^{-1}	μ /Pa s	k /W m^{-1} K^{-1}	L_H /kJ kg^{-1}	β /K	T_m /°C	Material	C_p /J kg^{-1} K^{-1}	k /W m^{-1} K^{-1}	ρ /kg m^{-3}		
Parida et al. [127]	Paraffin	805	2.10	3.65×10^{-3}	0.22	–	3.08×10^{-4}	333	Aluminum	910	237	2830	Numerical	Buoyant convection is prominent with greater porosity and larger pore size
Deng et al. [128]	Lauric acid	869	2.15/2.203	–	0.15/ 0.22	178		316	Aluminum	871	202.4	2719	Numerical	Porous metal foam leads to lengthier melting periods and lesser temperature dependence
Dinesh et al. [129]	Paraffin	880	2.10	–	0.2	169	–	335	Aluminum	910	205	2830	Numerical	Thermal performance influenced by foam geometry in PCM-metal foam-based energy storage units
Baruah et al. [130]	Paraffin	805	2.10	3.65×10^{-3}	0.22	120	3.08×10^{-4}	333	Aluminum	910	237	2830	Numerical	Lower porosity foam melts more quickly, while higher porosity foam can hold more latent energy
Jiao et al. [131]	OP42E paraffin	913/ 784	2.701	–	0.26/0.18	12.538/ 176.333	–	304.6/316.8	Copper/ Nickel	385/ 440	401.0/ 90.7	8960/ 8902	Numerical	Mixture theory model helps solve melting phenomena of pure PCM and PCM/ metal foam composite

Table 4 (continued)

References	Used PCM	Property of PCM							Property of metal foam/ Heat pipe/ Nano-particle				Case study	Important findings
		ρ /kg m^{-3}	C_p /kJ kg^{-1} K^{-1}	μ /Pa s	k /W m^{-1} K^{-1}	L_H /kJ kg^{-1}	β /K	T_m /°C	Material	C_p /J kg^{-1} K^{-1}	k /W m^{-1} K^{-1}	ρ /kg m^{-3}		
Esapour et al. [132]	RT 35	860/770	2.00	0.023	0.2	160	6×10^{-4}	302/ 309	Copper	380	400	8920	Numerical	Inner tube arrangement in metal foam/PCM composite significantly impacts internal phase change
Huang et al. [133]	RT 42	760	2.00	2.3×10^{-3}	0.2	174	6×10^{-4}	311–316	Copper	381	387.6	8978	Numerical	Natural convection may not always favor JPCM/ metal foam composite melting
Sardari et al. [134]	RT 54HC	850/800	2.00	3.65×10^{-3}	0.2	200	3.08×10^{-4}	326/ 327	Aluminum				–	Porous-PCM CLHS alternative maintains an almost consistent temperature of 54 °C during discharge operation
Iasiello et al. [135]	Paraffin (Ph EUR, BP, NF CAS 8002–74-2)	850	2.49	3.85×10^{-3}	0.15/ 0.20	185	7.78×10^{-4}	350	Aluminum	900	220	2700	Experimental and Numerical	Liquid phase convection significantly impacts the melting front position

Table 4 (continued)

References	Used PCM	Property of PCM							Property of metal foam/ Heat pipe/ Nano-particle				Case study	Important findings
		ρ /kg m^{-3}	C_p /kJ kg^{-1} K^{-1}	μ /Pa s	k /W m^{-1} K^{-1}	L_H /kJ kg^{-1}	β /K	T_m /°C	Material	C_p /J kg^{-1} K^{-1}	k /W m^{-1} K^{-1}	ρ /kg m^{-3}		
Mahdi et al. [136]	RT70HC	880/770	2.00	5.6×10^{-3}	0.2	260	10×10^{-4}	342/ 344					Numerical	The presence of porous material ensures a constant temperature profile, enabling extended charge and discharge periods

conducted a numerical study on PCM cement mortar, suggesting that encapsulated PCMs with improved thermal insulation could be suitable for food packaging. Hoang et al. [148] found that PCM (RT5) coated with polycaprolactone (PCL) demonstrated improved thermal buffering compared to regular PCM. Joulin et al. [149] explored thermal classification methods for building materials and identified microencapsulated PCMs as promising for thermal energy storage in structures. Rhafiki et al. [150] analyzed wallboards with microencapsulated PCM, showing their potential for thermal regulation. Cui et al. [151] used hollow steel spheres (HSB) to encapsulate PCM in concrete, resulting in better thermal–mechanical properties and higher heat cycle durability. Strengths of 50%-HSBC-c and 75%-HSBC-c increased by 14% and 22% after 90 heat cycles. Yan et al. [152] developed a PCM wall with pipe-encapsulated devices for self-triggered thermal energy release, resulting in a 74.5% reduction in heat release compared to conventional walls in hot climates.

Gao et al. [153] reported on minerals encapsulated by PCM, providing insights into mineral-PCM interactions. Cheralathan et al. [154] experimentally studied a PCM-based CTES system merged with an industrial refrigeration system. They emphasized the advantages of charging storage devices at particular temperatures to improve thermal performance. Raj et al. [155] found that the shape of encapsulation significantly impacts the charging and discharging rates of organic paraffin wax PCM. They discovered that discrete encapsulated units lead to notable enhancements in operational efficiency. Grabo et al. [156] analyzed a TES unit with densely packed macro-encapsulated LHS components, showing higher packing density and energy storage efficiency. Efficiency increased from 51.5 to 64.1 kWh m^{-3}. Ouali et al. [157] studied temperature gradients in construction materials with microencapsulated PCM, revealing thermal insights.

Table 6 summarizes research paper reviews and thermophysical characteristics of encapsulated PCMs for enhancing LHTES systems. Encapsulation methods play a crucial role in enhancing the stability, thermal performance, and usability of PCMs in various applications. Microencapsulation techniques, as demonstrated by Tittelein et al. [147] and Hoang et al. [148], involve coating the PCM with protective shells that improve its thermal cycling stability and prevent leakage. For example, Hoang et al. [148] observed that encapsulated PCMs have superior thermal storage capacity, which is particularly useful for thermal shielding in perishable items. Macro-encapsulation, which involves containing PCMs in larger shells or containers, is effective for large-scale applications. However, studies such as Joulin et al. [149] show that microencapsulated PCMs can store and release energy at a rate 41% higher than conventional mortar, indicating a significant performance advantage. Rhafiki et al. [150] further highlight that heat flux measurements on both sides of encapsulated materials can rapidly ascertain their thermophysical properties, suggesting encapsulated PCMs and its suitability for applications requiring precise thermal management.

The microencapsulation enhances thermal reliability but can increase material costs, while macro-encapsulation offers better scalability but may suffer from lower heat transfer efficiency. Therefore, selecting an appropriate encapsulation method depends on the application requirements, such as stability, efficiency, and cost.

Energy storage with nano-enhanced PCM (NePCM)

NePCMs integrate nanoparticles into conventional PCMs to improve thermal properties. NePCMs improve PCM performance by dispersing nanoparticles (< 100 nm), such as carbon-based materials (carbon, graphite, grapheme, CNTs), base metals (Ag, Cu), and metal oxides (CuO, Al_2O_3, MgO, TiO_2, ZnO), enhancing thermal conductivity [158–160]. Various conductive nanomaterials come in different shapes: 0D-like diamond particles, 1D-like fibers and tubes, and 2D-like sheets. Mixing them with PCMs creates colloidal NePCM, used in thermal systems. In fact, there are several influence of nanoparticles on the heat transfer mechanisms in NePCMs.

The presence of high-conductivity nanoparticles—alters the fundamental heat transfer mechanisms through Brownian motion, clustering effects, and thermal percolation. These mechanisms collectively enhance thermal conductivity, improve energy storage, and optimize heat dissipation within the PCM matrix.

(a) Brownian motion and its role in microconvection

Brownian motion describes the spontaneous, erratic movement of nanoparticles in a fluid due to thermal energy fluctuations. This random motion induces microconvection, which serves as an additional heat transfer pathway alongside conduction. The main consequences include:

- Moving nanoparticles act as carriers of heat, redistributing thermal energy more efficiently within the PCM and reducing temperature gradients.
- The dynamic motion of nanoparticles causes localized turbulence, leading to an enhancement in overall heat conduction.
- Smaller nanoparticles exhibit stronger Brownian motion, amplifying microconvection effects. However, if the concentration is too high, nanoparticles tend to aggregate, which counteracts the benefits.

While Brownian motion is highly effective in the liquid phase of PCMs, its influence becomes less significant in the solid phase due to restricted mobility.

(b) Clustering effects

Nanoparticles suspended in NePCMs tend to form clusters due to intermolecular forces, particularly Van der Waals attractions. Clustering impacts thermal performance in two contrasting ways:

- When clusters form well-structured networks, they act as thermal bridges, facilitating faster heat conduction. This effect is pronounced when high-conductivity materials like metallic or graphene-based nanoparticles aggregate in an ordered manner.
- Excessive clustering leads to agglomeration and sedimentation, disrupting heat transfer uniformity. Instead of enhancing conductivity, these clusters introduce resistance, impeding effective thermal transport.

The extent of clustering depends on factors such as nanoparticle loading, surface properties, and dispersion stability. Employing surfactants or functionalized nanoparticles helps prevent excessive aggregation, ensuring an optimal balance between improved conductivity and stability [161].

(c) Thermal percolation and its impact on conductivity

Thermal percolation occurs when a critical nanoparticle concentration is reached, forming an interconnected network that significantly enhances heat conduction. The key aspects of this phenomenon include:

- Beyond a specific volume fraction, nanoparticles establish a continuous conductive network, leading to a substantial increase in thermal conductivity. This effect is most prominent in materials with high aspect ratios, such as CNTs and graphene nanoplatelets.
- In solid-state NePCMs, percolation networks provide efficient pathways for phonon propagation, minimizing thermal resistance and accelerating heat dissipation.
- While increasing nanoparticle concentration enhances percolation, an excessive amount can result in sedimentation or phase separation, reducing long-term reliability.

To maximize performance, the selection of nanoparticle type, concentration, and dispersion strategy must be carefully optimized to achieve effective percolation while maintaining stability. Therefore, the thermal performance of NePCMs is dictated by the interplay of Brownian motion, clustering dynamics, and percolation effects. While

Table 5 A summary of studies of PCM with heat pipe

References	Used PCM	Property of PCM							Property of metal foam/Heat pipe/nanoparticle				Case Study	Important findings
		ρ /kg m^{-3}	C_p /kJ kg^{-1} K^{-1}	μ /Pa s	k /W m^{-1} K^{-1}	L_H /kJ kg^{-1}	β /K	T_m /°C	Material	C_p /J kg^{-1} K^{-1}	k /W m^{-1} K^{-1}	ρ /kg m^{-3}		
Sharif et al. [137]	Sodium nitrate	2000	1.73	3.02×10^{-3}	0.57	182	6.6×10^{-4}	580	304 Stainless steel	557	19.8	7900	Numerical	Melting rates increase with heat pipe diameter or length of the condenser
Sharif et al. [138]	n-Octadecane	800/770	1.912/0.216	3.09×10^{-3}	0.358/0.148	243.5	9×10^{-4}	301	Copper	385	401	8933	Experimental and Numerical	Melting rates of HP-Foil structures are about nine times higher than those of bare copper rods with heat pipe-equivalent diameters
Xiaohong et al. [139]	LiF	2330/1800	2.35/2.45	2.23×10^{-5}	6.2/1.7	1040	–	1121	Sodium				Numerical	Large temperature difference in PCM region due to empty cavity, impacting heat pipe lifespan of the receiver and thermal stress
Shabgard et al. [140]	KNO_3	2109	0.953	2.59×10^{-3}	0.5/0.425	95	200×10^{-6}	608	Stainless steel	2.89×10^{6}	15			Dimensionless heat pipe efficacy increases with more heat pipes
Peng et al. [141]	Paraffin wax	880/770	2.15/2.25	–	0.21/0.18	245	–	303–313	Aluminum	871	202.4	2719	Numerical	PCM liquid portion distribution is not constant during discharge
Tiari et al. [142]	Rubitherm RT55	880/770	2.00	–	0.2	172 ± 7.5%	–	324–330	Copper	–	–	–	Experimental	Charging time is influenced markedly by HTF mass flow during charging

Table 5 (continued)

References	Used PCM	Property of PCM							Property of metal foam/Heat pipe/nanoparticle				Case Study	Important findings
		ρ /kg m^{-3}	C_p /kJ kg^{-1} K^{-1}	μ /Pa s	k /W m^{-1} K^{-1}	L_H /kJ kg^{-1}	β /K	T_m /°C	Material	C_p /J kg^{-1} K^{-1}	k /W m^{-1} K^{-1}	ρ /kg m^{-3}		
Pawar et al. [143]	PCM-72 (Tritriacontane paraffin, C33H68)	810/ 765	0.87/ 1.11	26×10^{-3}	0.21	256	–	338.83/ 345.18	Copper	381	38,000	8978	Numerical	PCM module shows the highest temperature difference of 30 °C during the evening rush
Chen et al. [144]	Paraffin wax	950	3.00	–	7.654	141.7	–	315.15–317.15	Copper	381	2000	8978	Numerical	Changing PCM thickness distribution greatly improves system performance
Behi et al. [145]	RT42	0.88/ 0.76	2.00	–	0.2	–	–	311–316	Copper				Experimental and Numerical	PCM-assisted heat pipes can meet up to 86.7% of the cooling load for a 50–80W power range
Behi et al. [146]	Organic paraffin wax	0.8/ 0.85	2.50	–	0.2/0.4	220	–	298–305	–		8212		Experimental and Numerical	Heat pipe and PCM-integrated heat pipe reduce the highest cell temperature by up to 17.3% and 40.7%, respectively

Table 6 A summary of studies of various encapsulated PCM

References	Used PCM	Property of encapsulated PCM							Case study	Important findings
		ρ /kg m^{-3}	C_p /kJ kg^{-1} K^{-1}	μ /Pa s	k /W m^{-1} K^{-1}	L_H /kJ kg^{-1}	β /K	T_m /°C		
Tittelein et al. [147]	Micronal PCM, DS 5001 X	1412	1.120 ± 0.060/1.080 ± 0.060	–	0.55 ± 0.02	11,600 ± 1160	–	299	Experimental	The enthalpy approach yields better outcomes compared to the apparent specific capacity approach
Hoang et al. [148]	RT 5	336/520	1.701	–	0.08	–	–	–	Experimental	Encapsulated PCM material has superior thermal storage capacity, useful for a thermal shield of perishable items
Joulin et al. [149]	Micronal PCM DS 5001 X	–	1.255 ± 0.063	–	0.37 ± 0.0	19,520 ± 1952	–	–	Experimental	PCM stores and releases energy at a 41% higher rate than conventional mortar in the same temperature range
Rhafiki et al. [150]		995	1.70/2.153	–	0.2/0.13	16,674	–	299	Numerical	Heat flux curves on both sides of the material facilitate rapid ascertainment of its thermophysical properties
Cui et al. [151]	PCM-HSBC-c	780	–	–	–	249,700	–	300.1	Experimental	50%-HSBC-c and 75%-HSBC-c strengths increase by 14% and 22%, respectively, after 90 heat cycles
Yan et al. [152]	RT28	750	1.981	–	9.5/9.8	160.2	–	299.5–301.5	Experimental	Pipe-encapsulated PCM wall reduces nighttime heat release by 74.5% compared to conventional walls
Cheralathan et al. [154]		917/1000	2.01/4.18	–	2.22/0.566	333.6	–	–	Experimental	The thermal performance of the storage system improves by charging at ideal evaporator temperatures
Raj et al. [155]	Paraffin wax	893/795	2.10/2.52	–	0.241/0.156	180.80	–	331–333	Experimental	The operational time of dual-pass solar air heaters is slightly reduced with a marginal increase in construction costs.

Table 6 (continued)

References	Used PCM	Property of encapsulated PCM							Case study	Important findings
		ρ /kg m^{-3}	C_p /kJ kg^{-1} K^{-1}	μ /Pa s	k /W m^{-1} K^{-1}	L_H /kJ kg^{-1}	β /K	T_m /°C		
Grabo et al. [156]		1280	3.00	–	1/0.6	240	–	–	Numerical	Stored energy efficiency increases from 51.5 to 64.1 kWh m^{-3}
Ouali et al. [157]		995	1.70/2.153		0.2/0.13	166.74		299	Experimental	Specific enthalpy, apparent heat capacity, and equivalent heat capacity magnitudes could be overstated during heating or cooling operations

Brownian motion enhances microconvection, clustering can either facilitate or obstruct heat transfer, depending on the extent of aggregation. The formation of percolation networks significantly boosts thermal conductivity but requires careful control to avoid instability. By fine-tuning nanoparticle parameters, NePCMs can be tailored for superior heat transfer performance, making them promising candidates for advanced thermal energy storage applications [162, 163].

Several studies are ongoing to enhance NePCM properties and stability [164, 165]. Moreover, several researchers have conducted reviews to summarize existing NePCM works and outline future research directions [8, 166]. Key determinants for thermal enhancement include the base PCM category and the type, form, size, and concentration of nanoparticles [167]. Experimental research on NePCM in TES systems is limited. Table 7 summarizes notable investigations and enhancements.

At various temperatures, different nanomaterials have been incorporated into base PCMs to enhance thermal properties. For instance, at − 9.3 °C, titanium dioxide nanoparticles (20 nm) in barium chloride hydrate achieve a 13% increase in thermal conductivity [168]. At − 0.8–0 °C, copper oxide nanoparticles (37–59 nm) in water improve solidification rates by 35% [169]. In the 5.8–7.7 °C range, silver nanoparticles (10–18 nm) in organic ester boost thermal conductivity by 67% [170]. Copper nanowires (50 nm diameter, 10 µm length) in calcium chloride at 28–30 °C enhance thermal conductivity by 52% [171], while graphite nanoplatelets (15 µm diameter, 35 nm thickness) in paraffin wax at 29.7 °C achieve a remarkable 167% increase in thermal conductivity [172]. At higher temperatures, carbon nanotubes (6–9 nm diameter, 5 µm length) in N-eicosane show a 113% improvement in thermal conductivity at 32–40 °C [173], and graphene nanoplatelets (5–10 µm diameter, 4–20 nm thickness) in N-eicosane achieve a 400% enhancement at 35.7 °C [174]. Copper nanowires (80–150 nm diameter, 10 µm length) in tetradecanol exhibit an astonishing 793% increase in thermal conductivity at 38–40 °C [175]. Magnetite nanoparticles (40–75 nm) in paraffin wax at 43–50 °C increase thermal conductivity by 60% [176]. Silver nanowires (50–100 nm diameters, 5–20 µm length) in polyethylene glycol achieve a significant 1130% improvement at 43–60 °C [177]. In the range of 51–55 °C, graphite nanoplatelets (15 µm diameter, 10 nm thickness) in paraffin wax result in a 207% increase in thermal conductivity [178]. Boron nitride nanosheets (0.5–2 µm length, 100 nm thickness) in paraffin wax enhance thermal conductivity by 60% at 50–58 °C [179], while graphene nanoplatelets (10 µm diameter, 64 nm thickness) in paraffin wax improve thermal conductivity by 164% at 51–58 °C [180]. Titanium dioxide nanoparticles (50 nm) in paraffin wax at 55–58 °C achieve a 25% increase in thermal conductivity [181], and copper nanoparticles (25 nm) in paraffin wax enhance melting rates by

30% at 55–60 °C [182]. Sodium acetate trihydrate with copper nanoparticles (10–30 nm) increases thermal conductivity by 25% at 58–62 °C [183], while palmitic acid with titanium dioxide nanoparticles (21 nm) shows an 80% improvement in thermal conductivity at 60–62 °C [184]. Lastly, alumina nanoparticles (10–20 nm) in paraffin wax result in a 43% increase in thermal conductivity at 61.6 °C [185].

Other studies have explored using NePCMs to improve thermal energy storage systems. Sheikholeslami et al. [186] simulated PCM charging with nanoparticles, discovering higher nanoparticle concentration speeds up PCM melting. Algarni et al. [187] developed a new evacuated tube solar collector (ETSC) with a NePCM-based LHTES system, enhancing its efficiency by 32% with just 0.33% added mass of PCM/copper composite. Sheikholeslami [188] found that nanoparticle size, volume fraction, and other factors greatly affect solidification enhancement in NePCM. Khodadadi and Hosseinizadeh [189] found that NePCM has higher thermal conductivity than traditional PCM, making it promising for diverse energy applications. Sheikholeslami [190] found that platelet-shaped nanoparticles speed up solidification. Hosseinizadeh et al. [191] used RT27 PCM and copper nanoparticles to study NePCM melting in a spherical vessel. Their findings showed that nanoparticles enhance thermal conductivity, resulting in quicker melting times. Babazadeh et al. [192] simulated CuO-water solidification in a heat-release unit with new fins. They discovered that nanoparticles speed up PCM solidification, with a 40 nm diameter being optimal for enhancing discharge rate. Sheikholeslami [193] found that the solidification time of a magnetic field affected NePCM is inversely related to Hartmann number and nanofluid concentration. Sheikholeslami [194] simulated Lorentz forces that impact on heat transport in porous NePCM during solidification, discovering they boost flow rate and cut discharge time. Kashani et al. [195] examined the impacts of surface undulation (0, 0.125, 0.25, and 0.5) and nanoparticle concentration ($\varphi = 0, 0.05, 0.1$) on freezing of Cu-water nanofluid within a vertical container undergoing buoyant convection (Grashof numbers, $Gr = 10^5, 10^6, 10^7$) and found that surface waviness can regulate solidification time. Abu-Hamdeh et al. [196] found that in a parabolic trough collector with oil-Ne-PCM/Al2O3, heat transport increases by up to 30% with higher Rayleigh and Hartman numbers. These studies add to the research on nano-enhanced PCMs for better thermal energy storage [197]. Refer to Table 8 for details on PCM and nanoparticle properties in this research. Synthesis of NePCM.

NePCMs are advanced materials used in latent heat energy storage units to overcome the limitations of conventional PCMs. Traditional PCMs store and release thermal energy during phase transitions, such as melting and solidifying, but often suffer from low thermal conductivity, limiting their efficiency. NePCM represents the future of energy storage systems due to their superior thermal management properties. They offer high thermal conductivity, increased energy density, rapid thermal response, enhanced stability and longevity, sustainability, and versatility in applications ranging from solar power plants and automotive systems to aerospace and self-powering sensors. This adaptability enhances their potential impact across various industries. These are key components in developing next-generation energy storage systems. Their ability to improve efficiency and reliability across various applications highlights their potential to drive innovation in energy storage technology. Based on the table summarizing various studies on NePCMs, carbon-based nanoparticles (e.g., CuO, Al_2O_3, and copper) have been found to significantly impact the thermal properties of PCMs. For instance, Sheikholeslami et al. [186] reported an increase in the melt fraction with the rising concentration of CuO nanoparticles in paraffin wax. Similarly, Khodadadi et al. [189] found that adding copper nanoparticles to water increased both the thermal conductivity and heat-release rate, making the NePCM more suitable for energy-related applications.

Notably, the choice of nanoparticle type also influences the optimal conditions for phase change. For example, Sheikholeslami [188] highlighted that using CuO nanoparticles in water accelerates PCM solidification, with the 40 nm diameter being the most effective. While carbon-based nanoparticles such as CuO and Al_2O_3 show significant thermal enhancement, cost and environmental concerns need to be considered for large-scale applications. In addition, studies by Abu-Hamdeh et al. [196] indicate that using Al_2O_3 nanoparticles results in a lower receiver temperature in oil-based PCMs compared to other NePCMs, pointing toward its potential for specific applications.

Thus, while carbon-based nanoparticles enhance thermal conductivity and accelerate phase change processes, their selection must consider the trade-offs in cost, stability, and environmental impact. Although in NePCMs, incorporating nanoparticles at concentrations exceeding 3–5% often becomes impractical due to several challenges. Higher nanoparticle concentrations can lead to significant increases in viscosity, which hampers the flow and uniform distribution of the PCM, making it difficult to achieve desired thermal performance. Additionally, excessive nanoparticles can cause aggregation, reducing the effective surface area for thermal transfer and potentially impacting the stability and longevity of the PCM. Moreover, the cost of nanoparticles at high concentrations can be prohibitive, making it economically unfeasible for many applications. Consequently, while higher nanoparticle concentrations can theoretically enhance thermal properties, practical considerations often necessitate limiting their usage to 3–5% to balance performance and feasibility.

Table 7 An overview of experimental studies on NePCM and noticeable enhancement

PCM temperature	Morphology	Base PCM	Nanoparticles	average particle size	Concentration	Thermal enhancement	References
−9.3 °C	Nanoparticle	Barium chloride hydrate	Titanium dioxide	$d=20$ nm	1.13 vol%	13% thermal conductivity	[168]
−0.8–0 °C	Nanoparticle	Water	Copper oxide	$d=37–59$ nm	0.1 mass%	35% solidification rate	[169]
5.8–7.7 °C	Nanoparticle	Organic ester	Silver	$d=10–18$ nm	5.0 mass%	67% thermal conductivity	[170]
28–30 °C	Nanowire	Calcium chloride	Copper	$d=50$ nm, $l=10$ µm	0.17 mass%	52% thermal conductivity	[171]
29.7 °C	Nanoplatelet	Paraffin wax	Graphite	$d=15$ µm, $\delta=35$ nm	10.0 mass%	167% thermal conductivity	[172]
32–40 °C	Nanotube	N-eicosane	Carbon	$d=6–9$ nm, $l=5$ µm	1.0 mass%	113% thermal conductivity	[173]
35.7 °C	Nanoplatelet	N-eicosane	Graphene	$d=5–10$ µm, $\delta=4–20$ nm	10.0 mass%	400% thermal conductivity	[174]
38–40 °C	Nanowire	Tetradecanol Calcium chloride	Copper	$d=80–150$ nm, $l=10$ µm	11.9 vol%	793% thermal conductivity ↑	[175]
43–50 °C	Nanoparticle	Paraffin wax	Magnetite	$d=40–75$ nm	20.0 mass%	60% thermal conductivity	[176]
43–60 °C	Nanowire	Polyethylene glycol	Silver	$d=50–100$ nm, $l=5–20$ µm	19.3 mass%	1130% thermal conductivity	[177]
51–55 °C	Nanoplatelet	Paraffin wax	Graphite	$d=15\ \mu m$, $\delta=10\ nm$	7.0 mass%	207% thermal conductivity	[178]
50–58 °C	Nanosheet	Paraffin wax	Boron nitride	$l=0.5–2\ \mu m$, $\delta=100\ nm$	10.0 mass%	60% thermal conductivity	[179]
51–58 °C	Nanoplatelet	Paraffin wax	Graphene	$d=10\ \mu m$, $\delta=64\ nm$	5.0 mass%	164% thermal conductivity	[180]
55–58 °C	Nanoparticle	Paraffin wax	Titanium dioxide	*d of 50 nm*	0.3 mass%	25% thermal conductivity	[181]
55–60 °C	Nanoparticle	Paraffin wax	Copper	$d=25\ nm$	1.0 mass%	30% melting rate	[182]
58–62 °C	Nanoparticle	Sodium acetate trihydrate	Copper	$d=10–30\ nm$	0.5 mass%	25% thermal conductivity	[183]
60–62 °C	Nanoparticle	Palmitic acid	Titanium dioxide	$d=21\ nm$	5.0 mass%	80% thermal conductivity	[184]
61.6 °C	Nanoparticle	Paraffin wax	Alumina	$d=10–20\ nm$	10.0 mass%	43% thermal conductivity	[185]

The integration of various enhancement techniques, including fins, metal foams, heat pipes, and nanoparticles, significantly improves the thermal performance of PCMs across a wide range of applications. Fins, particularly those made from highly conductive materials like aluminum and copper, enhance heat transfer rates by as much as 155%, with optimized designs reducing charging and discharging times by up to 82.1%. The geometry and orientation of fins, such as rectangular, spider-web, and fractal designs, have a notable impact on melting and solidification times, making them ideal for applications requiring rapid heat storage and release.

Metal foams, especially aluminum and copper-based, further improve thermal conductivity by increasing the surface area and facilitating uniform heat distribution. While lower porosity metal foams lead to faster melting, higher porosity foams store more latent energy, making them valuable for energy storage applications that require both conductivity and prolonged heat release.

Heat pipes, when integrated with PCMs, exhibit remarkable efficiency in managing temperature gradients and increasing melting rates, sometimes by up to nine times compared to systems without heat pipes. Heat pipes made from copper or stainless steel help reduce thermal stresses, extend system life, and are particularly effective in concentrated solar power and cooling systems. The size, material, and condenser length of the heat pipes are critical factors in optimizing their performance. NePCMs, which incorporate nanoparticles such as CuO and Al_2O_3, improve thermal conductivity and speed up the phase change process. Even small concentrations of nanoparticles can lead to significant improvements in energy density and heat-release rates, making them effective in energy storage systems. However, the concentration of nanoparticles must be carefully optimized to prevent issues such as agglomeration that could impact system stability.

Combining PCMs with these enhancement techniques can greatly improve thermal performance, energy efficiency, and storage capacity. Each technique offers unique advantages tailored to specific applications, and future research should focus on optimizing material selection, geometrical design, and the synergistic effects of combining multiple enhancement methods for maximum performance, cost-efficiency, and long-term durability.

Synthesis of nano-enhanced PCM (NePCM)

Preparation of NePCM

NePCMs are synthesized by incorporating nanoparticles into base PCMs, requiring a clear understanding of both components' properties. Figure 7 illustrates the process. First, solid PCM and nanoparticles are precisely measured using a computerized balance scale. Then, the PCM is melted on a hot plate. Nanoparticles are slowly added to the molten PCM while stirring for about two hours at 500 RPM. Nanoparticles are evenly dispersed in the PCM through four hours of sonication with an ultrasonic vibrator. The temperature of vibrator must stay above the melting point of PCM to prevent solidification. Afterward, the NePCM is extracted and frozen for evaluation. Each experiment was repeated thrice to reduce variability.

Characterization of NePCM

NePCM characterization employs scanning electron microscope (SEM), transmission electron microscopy (TEM), X-ray diffraction (XRD), differential scanning calorimetry (DSC), field emission scanning electron microscope (FE-SEM), Fourier transform infrared spectroscopy (FT-IR), dynamic light scattering (DLS), and thermogravimetric analysis (TGA). SEM and FE-SEM examine morphology. TEM provides high-resolution imaging of nanoparticle dispersion. XRD analyzes crystal structure. Fourier transform infrared spectroscopy (FT-IR) studies the chemistry of solid surfaces and liquids, DLS measures nanoparticle dispersion in PCM, DSC is used to determine the melting point of NePCM, and TGA analyzes nanoparticle thermal stability.

Characteristics of NePCM

Studies consistently show that nanofluids have higher thermal conductivity than macro-particle suspensions. Metallic nanofluids typically show greater enhancement than oxide ones.

Influence of temperature on thermal conductivity of PCM/NePCM

As temperature rises, thermal conductivity increases nonlinearly due to accelerated nanoparticle mobility in NePCM [197, 198], driven by stronger Brownian forces. Nurdin and Satriananda [199] studied how temperature affects the thermal conductivity of maghemite nanofluids. They found that higher volume percentages lead to greater thermal conductivity, especially at higher temperatures. The most significant increase occurred at 60 °C with a volume concentration of 2.5% (Fig. 8a). Jiang et al. [200] used 20 nm copper nanoparticles to study how temperature affects thermal conductivity. They found that as temperature increased, thermal resistance unexpectedly rose. This increase was attributed to nano-convection induced by intensified Brownian motion of nanoparticles, resulting in more pronounced thermal conductivity compared to fluids with higher viscosity at 452.66 K. Águila et al. [201] experimentally found that using copper nanoparticles with octadecane as the base PCM resulted in a decrease in thermal conductivity as temperature increased, dropping from 0.147 to 0.131 W m^{-1} K^{-1}. Cieslinski et al. [202] found that adding 0.1% Al_2O_3 to water increased its thermal conductivity of 2.4% as the temperature rose from 20 to 40 °C. Additionally, Sedeh et al. [203] found that as the temperature increased, the thermal conductivity of CuO and TiO_2 nanoparticles rose, while that of the base fluid decreased. Patel et al. [204] also observed that a nanofluid's thermal conductivity significantly improves with increasing temperature.

Influence of nanoparticle size of NePCM

Nanoparticle critically influences NePCM thermal properties and stability, but its effect on thermal conductivity (k) differs in various studies. Keblinski et al. [205] found that despite short phonon mean free paths in fluids (1–2 nm), nanofluids have reduced inter-particle spacing, which facilitates ballistic phonon transport and enhances thermal conductivity. At 5% volume, 10 nm particles are 5 nm apart. Figure 8b shows how fluid (water) to nanofluid thermal conductivity varies with Al_2O_3 nanoparticles diameter. Tighter particle spacing due to enhanced Brownian motion improves heat transfer, suggesting ballistic heat transfer boosts thermal conductivity in nanofluids. Patel et al. [204] found that nanofluid thermal conductivity increases as particle size decreases.

Table 8 A summary of studies of PCM with nanoparticle

References	Used PCM	Property of PCM							Property of metal foam/ Heat pipe/ Nanoparticle				Case Study	Important findings
		ρ /kg m^{-3}	C_p /kJ kg^{-1} K^{-1}	μ /Pa s	k /W m^{-1} K^{-1}	L_H /kJ kg^{-1}	β /K^{-1}	T_m /°C	Material	C_p /J kg^{-1} K^{-1}	k /W m^{-1} K^{-1}	ρ /kg m^{-3}		
Sheikholeslami et al. [186]	Paraffin wax	865/ 770	1.934/ 2.196	3.85×10^{-3}	0.358/ 0.148	243.500	91×10^{-5}	301	CuO	540	18	6500	Numerical	Melt fraction increases with rising CuO nanoparticle concentration
Algarni et al. [187]	Paraffin wax	820	2.40	–	0.25	230	–	325	Copper	2380	0.31	837	Experimental	The presence of PCM enhances collector efficiency, leading to 32% overall performance augmentation
Sheikholeslami [188]	Water	997	4.179	–	0.6	335	–	–	CuO	540	18	6500	Numerical	Nanoparticle addition accelerates PCM solidification, with 40 nm diameter being optimal for solidification
Khodadadi et al. [189]	Water	997.1	4.179	8.9×10^{-4}	0.6	335	2.1×10^{-4}	–	Copper	383	400	8954	Numerical	NePCM exhibits higher thermal conductivity and heat-release rate than standard PCM, suitable for energy-related industries
Sheikholeslami [190]	Water	997	4.179	–	0.6	335	–	–	CuO	540	18	6500	Numerical	Because of better thermal conductivity and higher heat-release rate of NEPCM in comparison with the conventional PCM, it has the potential to be used in more energy-related fields

Table 8 (continued)

References	Used PCM	Property of PCM							Property of metal foam/ Heat pipe/ Nanoparticle				Case Study	Important findings
		ρ /kg m^{-3}	C_p /kJ kg^{-1} K^{-1}	μ /Pa s	k /W m^{-1} K^{-1}	L_H /kJ kg^{-1}	β /K^{-1}	T_m /°C	Material	C_p /J kg^{-1} K^{-1}	k /W m^{-1} K^{-1}	ρ /kg m^{-3}		
Hosseinizadeh et al. [191]	RT27	870/760	2.40/1.80	3.42×10^{-3}	0.24/0.15	179	5×10^{-4}	301–303	Copper	383	400	8954	Numerical	NePCM melts faster than regular PCM as a result of intensification in thermal conducting property and a reduction in the latent heat of fusion
Babazadeh et al. [192]	Water	997	4.179	–	0.6	335	–	–	CuO	540	18	6500	Numerical	Nanoparticle concentrations impact both total energy and solid fraction of PCM
Sheikholeslami [193]	Water	997	4.179	–	0.6	335	–	–	CuO	540	18	6500	Numerical	Lorentz forces expedite PCM solidification, nanoparticle usage accelerates the solidification rate
Sheikholeslami [194]	Water	997	4.179	–	0.6	335	–	–	CuO	540	18	6500	Numerical	The rate of solidification is accelerated by the use of nanotechnology. As buoyancy force increases, solidification time increases
Kashani et al. [195]	Water	997.1	4.179	8.9×10^{-4}	2.3/0.6	335	2.1×10^{-4}	–	Cu	383	400	8954	Numerical	Surface waviness influences the solidification rate, with solidification time increasing with more wavelike patterns
Abu-Hamdeh et al. [196]	Oil	640	2.132	–	0.0733	–	6.4×10^{-4}	–	Al_2O_3	765	40	3970	Numerical	Using Al_2O_3 nanoparticles results in decreased receiver temperature compared to other NePCMs

Effect of viscosity of nanoparticle on NePCM

The nanoparticle content in PCM affects both heat storage and conductivity capabilities, but if the dynamic viscosity of the PCM increases, it can limit the dispersion of nanoparticles. Águila et al. [201] discovered viscosity decreases as temperature rises but increases with more nanoparticles. Nanoparticle addition changes the rheology of liquid from Newtonian to shear-thinning non-Newtonian, shown by viscosity tests at different deformation rates (30–80 RPM). Similarly, Cieslinski et al. [202] found viscosity drops as temperature rises. For instance, with 0.1% Al_2O_3 in water, viscosity decreased from 20 mPa to 9.5 mPa as temperature went from 20 to 40 °C. Ho and Gao [206] found that the viscosity of paraffin wax suspension with Al_2O_3 nanoparticles decreases as temperature rises, showing a nonlinear relationship. Jesumathy et al. [207] found that dynamic viscosity spiked in PCM with CuO nanoparticles when concentrations surpassed 10 mass percent. He et al. [168] explained that adding TiO_2 nanoparticles to $BaCl_2$ solution increases viscosity due to decreased particle distance at higher concentrations, resulting in stronger interactions between nanoparticles and water molecules.

Influence of nanoparticle concentration of NePCM

Higher particle concentration leads to greater enhancement of thermal conductivity in NePCM due to interactions among nanoparticles, interfacial layers, and the base PCM, driven by nanoparticle motion within the PCM. Nurdin and Satrianandа [199] found that as the volume fraction of particles increases in maghemite (γ-Fe_2O_3) nanofluids (base fluid water), thermal conductivity also increases, showing a nearly linear relationship. For example, at 0.5% particle concentration, thermal conductivity is 0.612 W m^{-1} K^{-1}, while at 2.5% concentration, it reaches 0.982 W m^{-1} K^{-1}. This indicates a rise in thermal conductivity with higher nanoparticle concentration. Refer to Fig. 8c for thermal conductivity trends with varying temperature and concentration [208].

Águila et al. [201] confirmed that higher nanoparticle concentration boosts both thermal conductivity and viscosity. Their study revealed a 9% increase in thermal conductivity compared to the carrier liquid, aligning with previous findings that higher nanoparticle concentration enhances thermal conductivity. However, Keblinski et al. [205] found that nanoparticle aggregation enhances thermal conductivity by creating lower resistance pathways, especially at higher concentrations. As aggregation increases, smaller particle clusters form in the carrier liquid, facilitating rapid heat transport and improving conductivity. Figure 8d illustrates this effect, showing how the volume of highly conducting clusters increases with aggregation. Consequently, thermal conductivity improves as the volume fraction declines [209, 210].

According to Sedeh et al. [203], the thermal conductivity of nanofluid rises with higher mass fractions of metal oxide nanoparticles. They noted negligible changes in thermal conductivity below 1% concentration but significant increases between 1 and 5% in molten paraffin. Similarly, Patel et al. [204] found that the thermal conductivity of suspended nanoparticles is higher at lower concentrations, suggesting a nonlinear relationship with particle concentration. Several researchers have reported applications of NePCM in diverse fields of thermal management and energy storage systems [211, 212].

Case studies for practical implementation

Considering the PCM as an energy storage medium, there are several practical applications. Table 9 presents various practical applications of PCMs across different energy storage systems. The primary focus lies in enhancing efficiency, reducing energy consumption, and improving cost-effectiveness across a wide range of sectors, including solar water heating, refrigeration, and building thermal regulation [213–215].

As in solar water heating systems, the use of water-based PCMs has extended operational times and increased energy efficiency by up to 30%, as seen in projects in Germany, Spain, and India. These improvements are facilitated through techniques such as sensible heat thermal energy storage (SHTES) and latent heat thermal energy storage (LHTES), which optimize energy storage and release. Similar enhancements are observed in community water heating systems, where PCMs contribute to peak load shaving and significant cost savings of up to 84.4%. In industrial cooling and refrigeration processes, PCM-based systems have led to

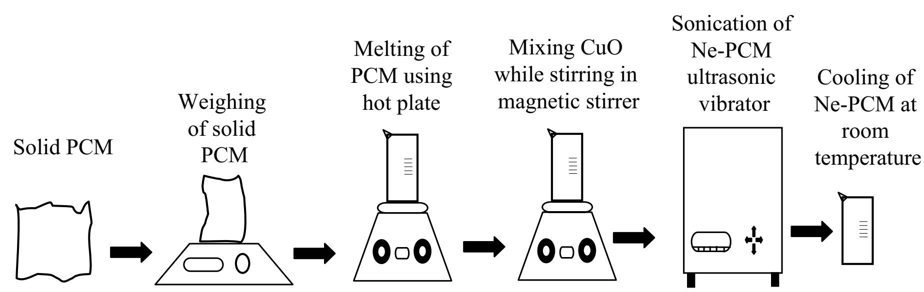


Fig. 7 Different steps for the preparation of NePCM

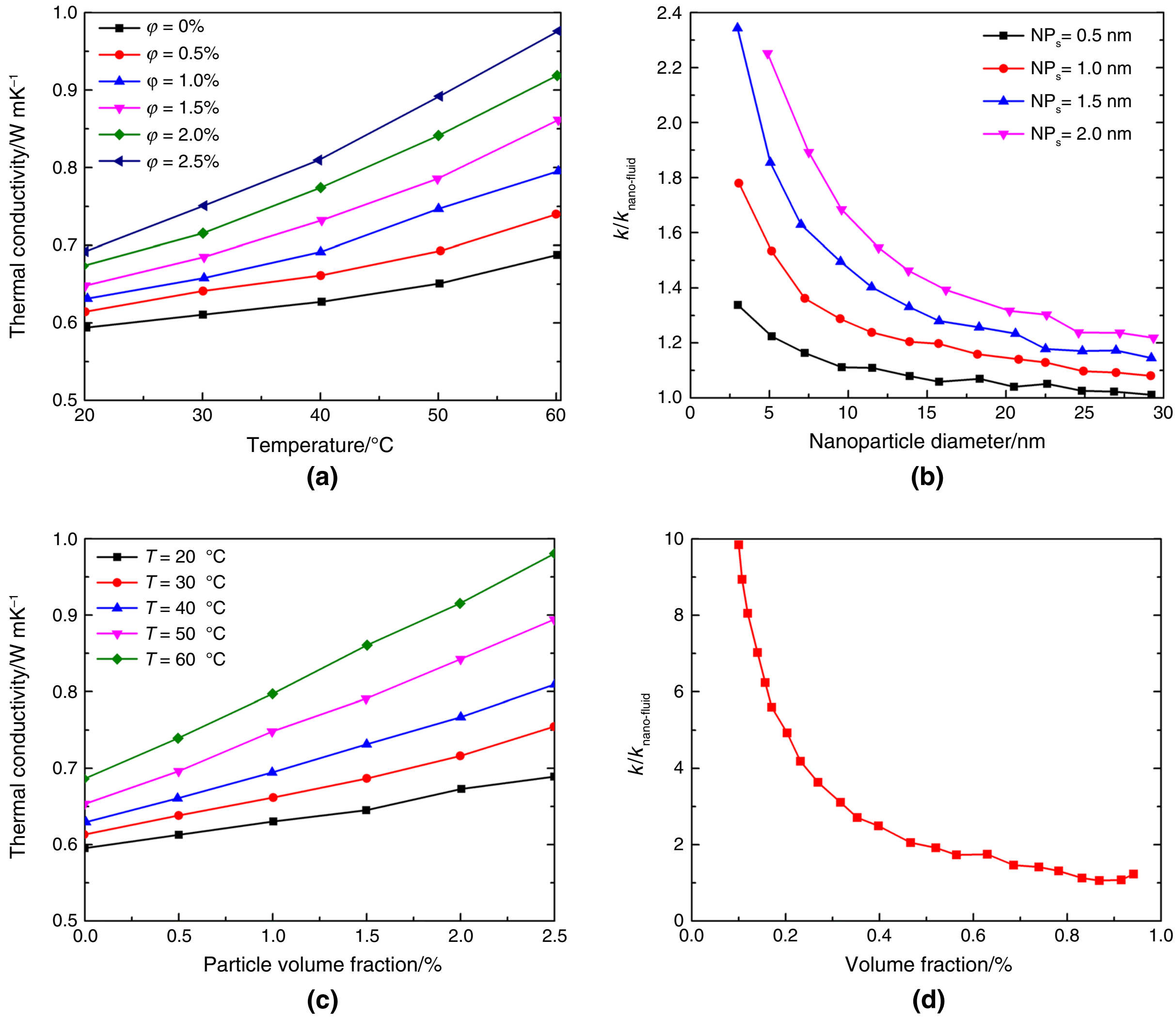


Fig. 8 Variation of **a** thermal conductivity (k) at different temperatures and nanoparticle volume fractions (φ) [199], **b** ratios of thermal conductivity of pure liquid and nanofluid for various nanoparticle sizes (NPs) [205], **c** thermal conductivity (k) at different nanoparticle concentrations (φ) and temperatures (T), and **d** ratio of thermal conductivity of pure fluid and nanofluid with respect to volume fraction (φ) [205]

substantial energy savings. A refrigeration system achieved a 40% reduction in lifecycle costs, while an industrial cooling system cut energy consumption by 17.1% and reduced CO_2 emissions by 17.5%. The integration of LHTES and advanced PCM technologies helped achieve these outcomes. In building applications, PCM implementations in energy-efficient projects reduced indoor temperature fluctuations by up to 30%, improving thermal comfort and energy efficiency. In the concentrated solar power (CSP) systems, PCMs like graphite and rocks have contributed to stabilizing discharge temperatures and increasing energy storage efficiency by 21.2%, as demonstrated in projects in the UAE and South Africa.

Thus, Table 9 shows how PCMs are versatile in enhancing energy efficiency across different applications and industries. By improving thermal regulation, energy storage density, and operational costs, these materials are proving to be valuable in the transition toward more sustainable energy solutions [216, 217].

Mathematical modeling

Standard governing equations, like those for heat transfer and fluid flow, are employed to model PCMs. The enthalpy-porosity approach is utilized to solve solidification and melting processes, treating the PCM as a porous medium. This approach adjusts the porosity from 0 (solid) to 1 (liquid) as the PCM undergoes phase transition. The enthalpy-porosity method has been widely used for modeling the melting and solidification processes in PCM-based systems. Voller and Prakash [229] and Brent et al. [230] demonstrated the application of this model in enhancing thermal energy storage. Recent studies have further explored the effectiveness of the enthalpy-porosity technique in simulating phase change behavior in various PCM configurations. These investigations provide valuable insights into optimizing PCM integration in thermal energy storage systems. By focusing on the enthalpy-porosity approach, we can gain a deeper understanding of its impact on the thermal efficiency of PCM-based energy storage units [231, 232]. Mathematical equations governing two-dimensional, incompressible, laminar flow are employed in the process [233].

Continuity equation

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \tag{1}$$

Momentum equation

$$\rho\left(\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}\right) = \mu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) - \frac{\partial p}{\partial x} - \frac{\mu}{k}u + A_{\text{mush}}\frac{(1-f)^2}{f^3+\delta}u \tag{2}$$

$$\rho\left(\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y}\right) = \mu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right) - \frac{\partial p}{\partial y} - \frac{\mu}{k}u + A_{\text{mush}}\frac{(1-f)^2}{f^3+\delta}v + \rho_{\text{ref}}g\beta(T - T_{\text{ref}}) \tag{3}$$

The Boussinesq approximation models buoyancy-driven convection by assuming constant fluid density in all equations except the body force term.

$$\left\{ f = \begin{array}{ll} f = 1, & T > T_{\text{m}} \\ \frac{T-T_{\text{s}}}{T_{\text{m}}-T_{\text{s}}}, & T_{\text{s}} < T < T_{\text{m}} \\ f = 0, & T < T_{\text{s}} \end{array} \right\} \tag{4}$$

These two $\frac{\mu}{k}u$ and $\frac{\mu}{k}v$ terms are the viscous loss term of the right side of Eqs. (2) and (3), which are added due to the porous model consideration. This part distinguishes the PCM area from the fin area in the momentum equation. Permeability (K) is set to 10^{-20} m^2 to prevent flow in the fin section, while $1/k$ is set to zero for the PCM section. A fourth term is added to the right side of Eqs. (2) and (3) in the enthalpy-porosity model as $A_{\text{mush}}\frac{(1-f)^2}{f^2+\delta}u$ and $A_{\text{mush}}\frac{(1-f)^2}{f^2+\delta}v$, where the liquid volume fraction is taken as f, and the mushy zone constant is taken as A_{mush}. In Eq. (4), f is described. For the numerical simulation, A_{mush} is set to 10^5 kg m^{-3} s^{-1}, as determined by an independent test. When $f = 0$, δ (0.001) is used to avoid division by zero. Natural convection is represented by the fifth term on the right side of Eq. (3).

Energy equation:

$$\left[(1-\varepsilon)\left(\rho C_{\text{p}}\right)_{\text{fin}} + \varepsilon\left(\rho C_{\text{p}}\right)_{\text{pcm}}\right]\frac{\partial T}{\partial t} + \varepsilon\rho_{\text{pcm}}L_{\text{H}}\frac{\partial f}{\partial t} + \left(\rho C_{\text{p}}\right)_{\text{pcm}}\left(u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y}\right) = \lambda_{\text{eff}}\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right) \tag{5}$$

In Eq. (5), ε represents porosity, with a value of 1 for the PCM region and 0 for the fin region. The right side of the equation for K_{eff} denotes the effective thermal conductivity, expressed as

$$K_{\text{eff}} = \varepsilon K_{\text{pcm}} + (1-\varepsilon)K_{\text{fin}} \tag{6}$$

Total enthalpy is the sum of sensible heat capacity and latent heat of phase change, defined as:

$$H = h + L_{\text{H}} \tag{7}$$

The sensible heat capacity is as follows:

$$h = h_{\text{ref}} + \int_{T_{\text{ref}}}^{T} C_{\text{p}}\,\text{d}T \tag{8}$$

Here, h_{ref} is the sensible enthalpy of PCM at T_{ref}.

In numerical simulations, both 2D and 3D models are employed to analyze PCMs, but 2D models are more frequently used by researchers. This preference for 2D models is largely due to their simplicity and reduced computational requirements compared to 3D models. 2D simulations are often sufficient for capturing the essential thermal behaviors and phase change characteristics of PCMs in many practical applications. They allow for quicker computations and easier implementation of boundary conditions and material properties. However, 3D models, while more complex and computationally demanding, provide a more detailed and accurate representation of PCM behavior, especially in scenarios involving intricate geometries or non-uniform boundary conditions. Notably, the difference between 2 and 3D models is typically less than 3%. Thus, 3D simulations are less commonly used primarily because of their higher computational cost and complexity. The 3D mathematical model is discussed in detail by Porto et al. [234]. The predicted thermophysical properties of the samples, based on the nanoparticle mass fraction, are evaluated using the mixture relations [235–239] summarized below

$$\text{Density}(\rho) \quad \rho_{\text{nano - pcm}} = (1-\varphi)\rho_{\text{pcm}} + \varphi\rho_{\text{nano - pcm}} \tag{9}$$

$$\text{Specific heat}\left(C_{\text{p}}\right) \quad C_{p_{\text{nano - pcm}}} = \frac{(1-\varphi)(\rho C_{\text{p}})_{pcm} + \varphi(\rho C_{\text{p}})_{\text{nanoparticle}}}{\rho_{\text{nano - particles}}} \tag{10}$$

Table 9 Summary of different types of energy storage systems and implementation of various applications

Medium (a)	Medium (b)	Applications	Improvements	Techniques	Case study	References
Water	PCM	Solar water heating	Extend the effective operational time	SHTES + LHTES	Solar water heating project	[218]
Water	PCM	Water heating system	Peak load shaving, energy saving up to 76.4%, spreading cost savings up to 84.4%	SHTES + LHTES	Community water heating system	[219]
Water	PCM	Solar water heating	System efficiency Improved substantially by 30%	LHTES	Solar thermal project	[220]
Water	PCM	Solar water heating	Energy storage density, overall cycle efficiency, and operation time increased up to by 30%, 16% and 25%, respectively	SHTES	Residential solar water system	[221]
Water	PCM	Electricity, water heating & storage	Enhancements in energy storage capacity and thermal management	LHTES	Hybrid solar power plant	[222]
Graphite	PCM	Concentrated solar tower	Energy storage efficiency improved by 21.2%	SHTES + PCM Integration	Solar power facility	[223]
Rocks	PCM	Concentrated solar tower	Stabilize the Discharge temperature	SHTES	Solar tower system	[224]
Bricks	PCM	Building enclosures	Indoor temperature fluctuations reduced by 30%	LHTES	Energy-efficient building	[225]
Roof	PCM	Building enclosures	Surface properties of roof materials enhanced to improve thermal comfort	LHTES	Green roof project	[226]
Water	Ice	Refrigeration process	40% reduction in the lifecycle cost	LHTES + LATEST	Supermarket cooling system	[227]
Ice	PCM	Refrigeration process	Reduction in energy consumption up to 17.1%, reduction in CO_2 emission up to 17.5%	LHTES + PCM Integration	Industrial cooling system	[228]

$$\text{Thermal conductivity}(K) \quad K_{\text{nano - pcm}} = K_{\text{pcm}}\left[\frac{\left(K_{\text{nanoparticles}} + 2K_{\text{pcm}}\right) - 2\varphi\left(K_{\text{pcm}} - K_{\text{nanoparticles}}\right)}{\left(K_{\text{nanoparticles}} + 2K_{\text{pcm}}\right) - \varphi\left(K_{\text{pcm}} - K_{\text{nanoparticles}}\right)}\right] \tag{11}$$

$$\text{Viscosity}(\mu) \quad \mu_{\text{nano - pcm}} = \frac{\mu_{\text{pcm}}}{(1-\varphi)^{2.5}} \tag{12}$$

$$\text{Thermal expansion coefficient } (\beta) \quad \beta_{\text{nano - pcm}} = \frac{(1-\varphi)(\rho\beta)_{\text{pcm}} + \varphi(\rho\beta)_{\text{nanoparticle}}}{\rho_{\text{nano - pcm}}} \tag{13}$$

$$\text{Latent heat}\left(L_{\text{H}}\right) \quad L_{\text{nano - pcm}} = \frac{(1-\varphi)(\rho L)_{\text{pcm}}}{\rho_{\text{nano - pcm}}} \tag{14}$$

Charging process:

$$\eta_{\text{charging}} = \frac{\int \dot{Q}_{\text{used}}\,\mathrm{d}t}{\int \dot{Q}_{\text{supplied}}\,\mathrm{d}t} = \frac{Q_{\text{used}}}{Q_{\text{supplied}}} \tag{15}$$

Discharging process:

$$\eta_{\text{discharging}} = \frac{\int \dot{Q}_{\text{used}}\,\mathrm{d}t}{\int \dot{Q}_{\text{stored}}\,\mathrm{d}t} = \frac{Q_{\text{used}}}{Q_{\text{stored}}} \tag{16}$$

Overall process:

$$\eta_{\text{charging}} = \frac{\int \dot{Q}_{\text{used}}\,\mathrm{d}t}{\int \dot{Q}_{\text{supplied}}\,\mathrm{d}t} = \frac{Q_{\text{used}}}{Q_{\text{supplied}}} \tag{17}$$

Energy transport performance is measured by the heat energy storage ratio, which compares the total stored energy in a thermal system to its maximum stored energy, and by the heat energy release factor.

$$\mathrm{Ratio}_{\text{heat storage}} = \frac{\int \dot{Q}_{\text{stored}} \mathrm{d}t}{\int \dot{Q}_{\text{maximum}} \mathrm{d}t} = \frac{Q_{\text{stored}}}{Q_{\text{maximum}}} \tag{18}$$

The heat-release ratio is the ratio of the total energy released from a thermal system to its highest contained energy.

$$\mathrm{Ratio}_{\text{heat release}} = \frac{\int \dot{Q}_{\text{released}} \mathrm{d}t}{\int \dot{Q}_{\text{maximum}} \mathrm{d}t} = \frac{Q_{\text{released}}}{Q_{\text{maximum}}} \tag{19}$$

During an energy analysis, only the quantity of energy delivered during charging and discharging can be determined, not the quality.

The following equations describe the heat transfer between the PCM zone and the fin.

$$Q = \frac{\lambda A_{\mathrm{t}}}{l_{\mathrm{t}}} \Delta t \tag{20}$$

The following equation describes the coefficient of thermal performance K_{t} of the PCM zone and fin zone.

$$K_{\mathrm{t}} = \frac{\lambda A_{\mathrm{t}}}{l_{\mathrm{t}}} \tag{21}$$

K_{t}: Coefficient of thermal performance of the fined and PCM area.

l_{t}: Maximum fin spacing, distance of thermal conduction between fin and PCM.

λ: PCM thermal conductivity.

A_{t}: Area of contact between fin and PCM.

Future scope

PCMs show great potential for use in LHTES systems. Various hybrid techniques, such as fins, geometry modification, metal foams, and nanoparticles, are being explored to improve their thermal behavior. Nanoparticle-based hybridization stands out as a particularly promising and straightforward method for enhancing LHTES efficiency.

This review highlights advancements in enhancing PCM thermal behavior, particularly through nanoparticle incorporation for better heat transfer and energy storage. Nano-enhanced PCMs show promise for cyclic thermal energy storage, despite stability and viscosity issues. Future research should focus on addressing these challenges:

(a) Investigating the optimal combination of thermal enhancement processes, such as extended surfaces (fins) combined with nanoparticles mixed with base PCM (NePCM), is crucial to addressing unresolved issues and ensuring long-term sustainability.
(b) Optimizing nanoparticle properties and loadings enhances phase change efficiency and heat storage, while preventing aggregation and sedimentation issues.
(c) Research on NePCM-based thermal energy storage systems should be expanded to optimize performance under varying conditions. Additionally, further studies on how nanoparticle characteristics affect NePCM performance are crucial.
(d) Numerical simulations are crucial for studying how nanoparticle characteristics affect melting and solidification. To accurately gauge system performance, factors like boundary conditions, TES geometries, thermal conductivity variations, and insertion size and shape must be incorporated in the models.
(e) Evaluate the trade-offs in NePCM systems, particularly how conduction-dominated heat transfer may reduce convective and latent heat transfer efficiency.
(f) Using nano-enhanced PCMs in thermal systems like electronics cooling, solar PV cooling, and space ventilation can help achieve ecological balance.
(g) Selecting nanostructures requires careful consideration of factors like thermal conductivity, aspect ratio, interface contact area, and viscosity changes.
(h) Implementing AI-based machine learning models to predict thermal properties and phase transition behavior more accurately [240]. Developing data-driven optimization approaches to fine-tune PCM formulations for improved performance. Machine learning techniques should be combined with experimental and computational studies to accelerate NePCM material development.
(i) Customizing enhanced PCMs to suit various applications, including thermal management in buildings, electronic cooling, and industrial waste heat recovery. Integrating PCM-based TES systems with renewable energy technologies such as solar and geothermal power. Conducting techno-economic assessments and life cycle analyses to determine the feasibility of large-scale PCM deployment.
(j) Exploring the potential of bio-derived nanomaterials such as cellulose nanofibers, chitosan, and lignin in NePCM development. Evaluating the biodegradability, recyclability, and environmental impact of PCM composites made from renewable sources [241, 242]. Enhancing the thermal and mechanical properties of

PCMs using bio-based reinforcements to improve sustainability.

Investigating new designs for compact TES containers using NePCM and fins—including central or peripheral placement of the heat transfer fluid—may reveal new opportunities for improving TES efficiency. This exploration and innovation will greatly boost the capabilities of hybrid TES units for sustainable energy storage.

Choosing the appropriate NePCM for a given application necessitates a thorough assessment of factors such as operating temperature, heat storage capacity, and long-term stability. The following guidelines provide a structured approach to identifying the best NePCM formulations based on specific application needs.

(a) Determining the required operating temperature range

- *Low-temperature applications (0–100 °C)* Ideal for thermal management in buildings, food storage, and electronic cooling. Organic PCMs, such as paraffin wax, are commonly used due to their high latent heat and chemical stability. To enhance thermal conductivity, carbon-based nanoparticles can be incorporated.
- *Medium-temperature applications (100–300 °C)* Suitable for industrial waste heat recovery, solar thermal storage, and automotive systems. Salt hydrates and fatty acids, when combined with metal oxide nanoparticles, improve thermal performance and durability.
- *High-temperature applications (> 300 °C)* Essential for concentrated solar power (CSP) plants and high-temperature industrial processes. Inorganic PCMs like molten salts or metal alloys, doped with ceramic or metallic nanoparticles, offer enhanced heat capacity and stability.

(b) Enhancing heat energy storage efficiency

- *Maximizing latent heat storage* Selecting PCMs with a high phase transition enthalpy ensures greater energy storage capacity. Nanomaterials such as graphene, carbon nanotubes (CNTs), and metallic oxides are effective in improving thermal conductivity without significantly affecting latent heat.
- *Optimizing nanoparticle concentration* Excessive nanoparticle loading can lead to aggregation, diminishing PCM effectiveness. Typically, an optimal range of 0.1–5 mass% is recommended to strike a balance between heat transfer enhancement and phase change efficiency.
- *Utilizing composite structures* Encapsulation and porous support materials, such as metal foams or aerogels, help prevent leakage while further improving heat transfer characteristics.

(c) Ensuring long-term stability and performance

- *Thermal cycling durability* The selected NePCM should retain its phase change characteristics over multiple heating and cooling cycles. The inclusion of stabilizers or surfactants can mitigate phase separation and performance degradation.
- *Chemical and structural integrity* Resistance to oxidation, minimal supercooling, and compatibility with containment materials are crucial for maintaining long-term reliability. Coatings with hybrid nanomaterials or encapsulated PCMs can enhance durability.
- *Minimizing leakage risks* Shape-stabilized PCMs, incorporating polymer matrices, aerogels, or porous carbon supports, help prevent leakage, improving practical applicability.

(d) Tailoring NePCMs to specific applications

- *Building and HVAC systems* Non-toxic, eco-friendly PCMs with moderate thermal conductivity and a phase transition temperature aligned with indoor conditions (20–35 °C).
- *Renewable energy storage (Solar TES, CSP)* High-temperature PCMs with superior thermal stability, often encapsulated to prevent degradation over time.
- *Electronics Thermal Management* PCMs with high thermal conductivity and rapid heat dissipation, reinforced with graphene-based nanomaterials.
- *Industrial waste heat recovery* Hybrid PCMs designed for broad temperature ranges with improved thermal conductivity to optimize heat absorption and discharge.

By carefully analyzing these factors, the most efficient and reliable NePCM formulation can be selected, ensuring optimal performance, longevity, and sustainability for specific thermal energy storage applications.

Conclusions

PCMs are crucial for boosting thermal energy storage (TES) system performance and advancing renewable energy storage. Particularly, nanoparticles possessing high thermal conductivity and low density are considered most effective.

Developing multifunctional nanocomposite PCMs balancing energy density and power is crucial. Exploring PCMs and nanoparticles together can enhance temperature control at a low cost. The key findings from this review are summarized below.

1. Enhancing PCM performance:
    - Nanoparticle inclusion speeds up melting/solidification, making nano-enhanced PCMs (NePCMs) great for thermal systems. They enhance phase transitions, boosting energy efficiency.
    - NePCMs significantly enhance the overall heat transfer and energy storage rates due to improved thermophysical properties imparted by nanoparticles.
2. Key observations:
    - *Choice of PCM* Different types of PCMs are optimal for specific temperature ranges and latent heat of fusion requirements. Salt-water eutectics, salt hydrates, and salts are preferred for low, medium, and high-temperature ranges, respectively.
    - *Thermal enhancement techniques* Fins and nanoparticles in NePCM boost thermal efficiency, striking a balance between performance and material consumption.
    - *Power density improvement* NePCMs enhance the power density rate of TES units despite higher viscosity, as conduction heat transfer predominates, minimizing the impact of natural convection intensity.
    - *Thermal conductivity* NePCMs exhibit superior thermal performance in TES and solar TES systems, with thermal conductivity influenced by nanoparticle volume concentration.
    - *Stability and uniformity* NePCM stability depends on PCM and nanoparticle selection, with surfactants aiding uniform nanoparticle dispersion.
    - *Temperature influence* Thermal conductivities typically increase nonlinearly with temperature due to increased nanoparticle mobility, though some cases may show conductivity reduction with temperature rise.
    - *Particle concentration* While increasing particle volume percentage generally enhances thermal conductivity, excessive concentrations may lead to diminishing returns due to particle aggregation.
    - *Molecular dynamics* Carbon-based nano-additives can be surface functionalized and doped to either counteract enthalpy decrease or boost thermal conductivity.
    - *Recommended nanoparticles* Carbon-based nanoparticles (GnP and CNT) and metal oxides in NePCM due to their great properties and performance.

Thus, NePCMs offer the potential to boost TES system efficiency and sustainability. Ongoing research targets nanoparticle selection, dispersion, and performance enhancement across diverse applications.

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